

THE PROLONGATION OF LIFE

DR. ALEXANDER A. BOGOMOLETS

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THE PROLONGATION OF LIFE

*The Robinson Foundation presents to the American Public this translation of Bogomolets' THE PRO-
LONGATION OF LIFE.*

Sooner or later every person becomes interested in the baffling problem of the secret of keeping young. That is why Dr. Bogomolets' new approach to the problem has received such widespread attention in America. Physicians have seen it abstracted in the American Review of Soviet Medicine, and the general reader has found it more popularly discussed in Martin Gumpert's book "You are Younger than you Think," in the Ladies' Home Journal, and in Reader's Digest.

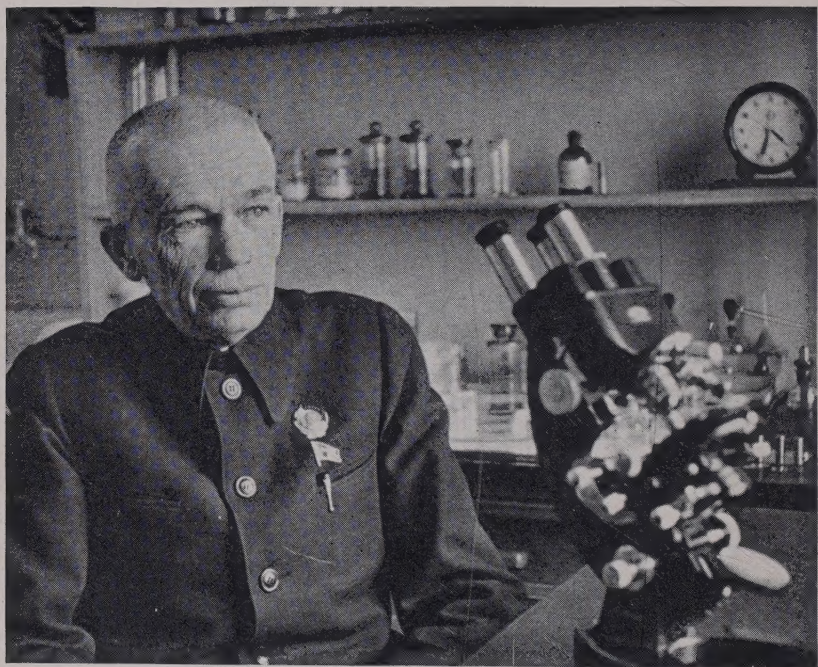
While these writings have been appearing, the present translation has been in process. The translators have spared no effort to produce a clear idiomatic text that faithfully reproduces the author's ideas and spirit. They have succeeded because of their unusual mastery of both Russian and American science and literature. To clear doubtful points they have communicated with Dr. Bogomolets.

Technical terms not found in the usual dictionary have been defined parenthetically or in footnotes, to assist the general reader.

In making available to non-Russian readers Bogomolets' startling approach to the problem of aging, The Robinson Foundation acts merely as a transmitter, without taking sides in controversial matters. The Foundation hopes that this book may be the first of many steps it will take to promote understanding of the problems of aging.

THE ROBINSON FOUNDATION, INC.

GEORGE P. ROBINSON, *President.*



Sovfoto

DR. ALEXANDER A. BOGOMELETS

The
PROLONGATION
of
LIFE

by

Dr. Alexander A. Bogomolets

*Translated by Peter V. Karpovich, M.D.,
and Sonia Bleeker*

A ROBINSON FOUNDATION PUBLICATION

ESSENTIAL BOOKS

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*To Dr. Oleg Bogomolets and his descendants
this book is dedicated*

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"I would like the anti-reticular cytotoxic serum, which proved so profitable to us, especially in the struggle with the after-effects of shock, to prove equally useful to our Allies. . . ."

(From a note to Prof. Irving Fisher, sent by the author together with the revised 1940 edition, October 20, 1944)

Preface

WITH the recent, tremendous development in means for the destruction of human life it is only natural that our attention should turn again to means for the prolongation of human life.

Most people dread old age and subsequent death. Death comes too soon.

From time immemorial man has tried to find ways for the eradication of physical suffering, for the restoration of youth, and for the prolongation of life. In the beginning, these means were mostly supernatural, created by dreaming and wishful thinking that brought about a rich folklore in which such agents as magic water could restore youth, heal wounds, and bring the dead back to life.

Gradually these fairy tale ideas have been harnessed and made to work in reality. Medicine has made great strides. Youth now may be restored in certain cases of endocrine deficiency; wounds can be healed quickly and efficiently; many apparently dead people are brought back to life; and the average span of man's life has been increased.

Until very recently our knowledge pertaining to the prolongation of life has depended mostly on haphazard gathering of "data" often of a dubious nature, and has had little of experimental background. At the present time, however, efforts are being made to apply methods

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common to all biological studies—experiments that should furnish objective information.

In this connection, the work of the Russian academician, Alexander A. Bogomolets, is of special interest. As Director of the Kiev Institute of Experimental Biology and Pathology he is in a position to test many ideas that hitherto have been either of a speculative nature or have been applied only to animals.

The present book is the result of Dr. Bogomolets' many years of work, study and thinking. It expresses his ideas, gives as much supporting data as he could muster, and outlines the problems that remain to be solved.

The translators have made every effort to preserve the spirit of the author's ideas and the manner of his presentation. The book contains debatable statements that will undoubtedly provoke discussion and may stimulate other investigators to undertake further research in problems related to the conservation and prolongation of human life.

PETER V. KARPOVICH, M.D.

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Alexander A. Bogomolets

DR. ALEXANDER ALEXANDROVICH BOGOMO-
lets, President of the Ukrainian Academy of Sciences,
has said, "It may sound paradoxical, but a man of 60 or
70 is still young. He has lived only half his natural life.
One can and must struggle against old age. It can be
treated just as any other illness because what we are
accustomed to regard as normal old age is actually an
abnormal, premature phenomenon."

Alexander Bogomolets was born in 1881, the son of a
young medical student awaiting trial for revolutionary
activity. She was held in Lubianovka prison on the out-
skirts of Kiev. After some time, the Tsarist court found
her guilty and exiled her to Siberia for life. Her son
was destined to become the father of longevity in the
Soviet Union, the country of his mother's dreams. He
was raised in the province of Poltava by his grandfather,
a retired officer of the army of Nicholas I.

In 1906, Bogomolets was graduated from the Univer-
sity of Odessa Medical School with high honors. He was
appointed to a position in the department of general
pathology. Five years later, in 1911, he went to the Uni-
versity of Saratov as professor of pathologic physiology.
With the advent of World War I, Bogomolets was as-
signed as consulting epidemiologist for the southwest-
ern front at Voronezh, a position he held throughout
the war and revolution.

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After the cessation of hostilities he resumed his research work at Moscow where he was professor of pathologic physiology at the Second Moscow University, from 1925 to 1930. Here he returned to his first interest, endocrinology. He had been a pupil of Metchnikoff and Ushinski and thus had inherited a passion for broad biologic problems and pathology. He was especially interested in the application of these branches of science to the fight against the deterioration caused by infectious diseases and old age. The results of his research and that of his students was summarized in a book *The Crisis of Endocrinology* which appeared in 1927.

Vegetative Centers of Metabolism was a result of his further studies on the nervous system. In this book he analyzed the complex problem of the nervous system and its relation to other organs, and advocated consideration of the organism as a unified whole.

Bogomolets, in 1924, had assisted in the founding of the Central Institute of Blood Transfusions, which did pioneer work in the field. It was at this institute that Bogomolets first evolved the theory of colloidoclasia to explain the therapeutic value of blood transfusions.

His theory of colloidoclasia attracted considerable attention at the first international congress on blood transfusion in Rome in 1935. Two years later, when the second congress was held in Paris, the majority of the papers presented took his theory as a point of departure. The exhibit of the work of his institute won a gold medal at the Paris World Exposition that year.

These various activities served as preparation for the real problem which engaged his thought—the problem of longevity. The Kiev Institute of Experimental

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Biology and Pathology was organized in 1930. Bogomolets was its founder and director. It was there that he proposed and developed the thesis of the active elements of connective tissue. This thesis is contained in his book, *Constitution and the Mesenchyme*. Under his direction, clinical studies by his 37 assistants indicated that the cells of the connective tissue intercept and destroy microbes which penetrate the body. The structure and condition of these cells determine the organism's resistance to infection, its predisposition to cancer and senile changes.

Bogomolets, having explained the reasons for premature senility, now set to work to discover the means by which the life span could be prolonged. On the basis of studies up to that time, the government increased the budget for the Institute of Experimental Biology and Pathology. The staff grew to 60. The institute became one of the finest in the country, fitted with the best obtainable equipment.

Careful research indicated that almost 30,000 persons in the Soviet Union were over 100 years of age. Bogomolets predicted that once the problem of longevity had been fully explored, this number would no longer be an index. The research continued. In 1938, an expedition went to Abkhasia in the Caucasus to examine 12 persons between the ages of 107 and 135. They found one, an extraordinarily healthy young man of 107 who confessed he planned to remarry.

During all his activity on the problem of longevity, Bogomolets found the time to direct the compilation of a *Guide to Pathological Physiology* for which, on March 13, 1941, he was awarded a Stalin Prize, first class.

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The results of the Abkhasian expedition culminated in the publication of the *Prolongation of Life*, in which Bogomolets advances the thesis that the usual life span of 50-80 years is the result of abnormal conditions and not nature's time limit for man. He draws the analogy that since the life span of an animal is 5-6 times longer than its period of maturation, therefore, the life span of man should be 125-150 years. Old age, according to Bogomolets, is influenced by social conditions, cold, hunger, poverty.

In addition to the post of President of the Ukrainian Academy of Sciences, Dr. Bogomolets is a Vice-President of the Academy of Sciences of the U.S.S.R. In 1937 he was elected deputy to the Supreme Soviet of the U.S.S.R., and in 1938, deputy to the Supreme Soviet of the Ukrainian S.S.R. He was awarded the Order of Lenin, the highest civilian decoration, for his scientific activity.

Dr. Bogomolets began to develop anti-reticular cytotoxic serum in order to put into practice the theory advanced in the *Prolongation of Life*. He experimented at first on animals, then on humans. A special clinic was established in his institute for the treatment of patients over 50. Small injections of the serum and special instructions on diet and hygiene produced satisfactory results. The health of the patients improved rapidly. Headaches, rheumatism, insomnia disappeared. The working capacity of the patients increased considerably.

The announcement of the perfection of the serum on June 23, 1941, coincided with the beginning of the war for the Soviet Union. The Institute of Experimental Biology and Pathology had to be evacuated from Kiev to Ufa. Here experiments on longevity were temporarily

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curtailed. The Institute's activities were directed toward producing enough serum to meet the needs of the battle-front. For, it had soon been demonstrated that ACS was effective in battle fractures, reducing the healing and bone-knitting time. ACS and Bogomolets became magic words in military hospitals throughout the country.

Two years ago, Dr. Bogomolets could proudly announce that the Ukrainian Academy of Sciences although evacuated deep into the hinterland, had carried on such good work that 80 papers on various subjects could be presented at the annual session. Specifically, the Ukrainian scientists had developed, in the course of the practical activity necessary during war time, new methods of treating wound infections; a new serum to accelerate the healing of wounds; perfection of methods of producing vitamin foods; a new system of plant nutrition, especially as applied to Kok-Sagiz, the rubber containing plants; as well as many practical applications of mathematical and physical principles to the production of aircraft, power development, new industrial processes.

Dr. Bogomolets is now 65. With the war ended, he looks forward to many years of intense activity, years he hopes that will bring us the solution of the problem of longevity.

(We are indebted to the American Review of Soviet Medicine for many of the facts in the above biographical sketch.)

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1. *Life and Death*

BIOLOGY, the study of life, pictures life as a continuous tide-like process consisting of building up and partial tearing down. In this process food particles are assimilated by the living organism; then they are broken down, thus liberating energy indispensable for life's functions. Various activities observed in the living organism may be explained on this basis. Growth, for instance, may be thought of as an excess of assimilation over dissimilation, that brings about the amassing of living matter in the cells. This in turn leads to growth beyond the individual limits, resulting in cell multiplication.

The primary characteristics of every living thing are the constancy of its chemical composition and shape, and the ability to reproduce "specific forms" of life, very similar to itself. Food and its assimilation lead to growth. But growth is limited by barriers nature has established over the individual cells. When the cell reaches its full growth, it divides into two cells. These two are in all respects similar to the mother-cell, and so begins the process of multiplication or growth beyond the limits of individuality. In the world of unicellular organisms, this process of life propagation goes on almost endlessly. Placed in a favorable environment, where accidental external causes of death are ruled out, these unicellular organisms create an impression of immortality.

2. *Cell Regeneration*

IN THE course of five years, Woodruff was fortunate enough to observe 3029 generations of infusoria resulting from successive cell division. Had most of these minute organisms not perished from unfavorable environment, their total mass would have outweighed the earth in less than five years; yet, there was no visible senility. Thus originated the erroneous idea, entertained by scientists for a long time, that in the early stages of its development life did not know the iron rule of death. Death, they thought, came as life evolved into the more complicated forms—those forms that the proud human brain considers to be the crown of nature.

However, who could distinguish mother from daughter in two individual infusoria that resulted from the division of the mother cell? The concept of a biological unit does not coincide at all with the understanding of the term "individual." An individual is something which cannot be divided. 2

The tremendous capacity for reparation and regeneration exhibited by living matter in these experiments is startling. A French scientist, Lumière, estimated that infusoria of the 40th generation may receive less than one trillionth part of the mother cell's substance. In Woodruff's experiments, with about three thousand divisions, the part of the mother cell's substance transmitted to the infusoria of the last generation could be

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represented by a fraction, having a numerator of one and a denominator containing one thousand digits.

Light travels at a speed of 186,000 miles per second and reaches the earth from Sirius, a nearby star, in the course of nine years. One has to compare a thousandth of a millimeter with this colossal distance, separating the earth from Sirius, to imagine the infinitesimal fraction of the original infusoria in Woodruff's first culture, that could still be present in any one descendant after 2,000 divisions.

It is understood that these astronomical figures are far beyond the possible division of living matter. But they give an idea of the ability of living matter to grow and reproduce.

Here is another illustration taken from Woodruff. An infant at the moment of its birth weighs 500 million times more than the fertilized egg cell. That increase in weight could be achieved by just 28 divisions of all the cells from the moment of conception—17 divisions in the first 15 days and 11 divisions in the course of the following $8\frac{1}{2}$ months.

3. *Records of Large Families*

BY COMPARISON with the productiveness of the human egg cell the productiveness of the human organism as a whole seems insignificant.

A. Bashutsky in his book, *St. Petersburg's Panorama*, published over 100 years ago, wrote:

"Here are two singular examples of the fertility of Russian women. In 1755 a peasant from the village of Vvedensky, Jacob Kirillov, was presented at Court. He was then 60 years old and had been married twice. His first wife bore 57 living children in 21 pregnancies, namely, four sets of quadruplets, seven sets of triplets and ten sets of twins—57 children in all. His second wife in seven pregnancies bore 15 children—one set of triplets and six sets of twins. Therefore, the old man had 72 children. On February 27, 1782, a report was received in Moscow from the Nikolsky monastery that a peasant, named Fedor Vasiliev, in Shuysk County, married twice, had 87 children by the two marriages. The first wife in 27 deliveries had four sets of quadruplets, seven sets of triplets, and 16 sets of twins—in all 69 children. The second wife had borne two sets of triplets and six sets of twins—18 children in all. Vasiliev was then 75 years of age and of the 87 offspring, 83 lived."

Remarkable as these examples of fertility are, they are insignificant in comparison with the usual rate of reproduction of the egg cell, whose product at the moment of an infant's birth exceeds the original weight by some hundred million times.

4. *Old Age as the Loss of Ability to Regenerate*

IF WE cut off the head part of a flatworm, a new head will grow fully replacing the old one. An earthworm, under similar conditions, will also grow a head, provided that the wound contains the end of the severed nerve. As the structure of the organism becomes more complicated and the function of the organs more differentiated, the ability to replace destroyed parts becomes more and more limited. Among the lower, cold-blooded vertebrates the regeneration of entire organs has also been observed (the tail, the fins, the extremities, and even an entire eye). This occurs if the viable embryonic cells, which represent the structural source of these organs, are left in place after the operation. Among the warm-blooded animals regeneration is possible only within the limits of certain tissues. Liver cells, for instance, can reproduce only liver cells.

However, even among the highest animals and man, the processes of regeneration go on during the entire lifetime. Despite the rhythmic succession of work and rest, the cells of the organs gradually wear out. To replace them, new generations of cells arise, born of the old cells and resembling them. The death and rebuilding of parts of the organs and tissues proceed continuously within the organism in the course of its entire life.

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The extent of these destructive and regenerative processes in the human organism might be visualized in the following illustration.

In the course of two months (some scientists think the period is somewhat longer) all the red blood corpuscles in man are replaced by new ones. This replacement coincides with the continuous destruction, mainly in the spleen, of some of the red blood corpuscles. The young red corpuscles arrive just as rapidly from the bone marrow, where they are formed as a result of the reproductions of certain cells called erythroblasts.

It is estimated that the adult man of average height has about four to five liters (4.2 to 5.3 quarts) of blood. A cubic millimeter of blood contains about five million red blood corpuscles. If we assume that one 60th of our red blood corpuscles is destroyed daily, it will mean the loss of 350-400 billion red blood corpuscles each day. Just as many, of course, are produced. This process goes on throughout life. Most remarkable is the fact that the products of decomposition of blood stimulate the formation of new blood.

Billions of cells are lost from the skin, whose upper layers peel off. Cells are lost when hairs are lost. Cells are lost during the process of continuous replacement of the mucous membranes of the respiratory, digestive, and urinary tracts. Cells are lost and are replaced by new ones in the internal organs. In the living organism enormous processes of destruction and construction go hand in hand without interruption.

One scientist, after counting the daily cellular losses of the organism and the corresponding renewal of tissues, came to the conclusion that in the course of seven

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years a man is entirely renewed, so that after seven years not one active, original cell remains. This conclusion, although entertaining, is basically false.

Not all human cells possess the same ability to multiply. Foremost are the cells that give origin to the formed elements of the blood, the cells that produce the spermatozoa, the skin epithelium, and the cells of connective tissue. Some cells of the internal organs, notably those of the liver, also possess the important ability to multiply and regenerate. The now commonly accepted concept is that after the birth of a child, the cells of its central nervous system do not multiply at all. One cannot help wondering at their longevity. After all, they have to take care of man's physical as well as psychic processes throughout the entire course of his long life. These are in fact the cells that are responsible for the maintenance of the unity of the personality and its development. The afore-mentioned scientist, who promoted the theory of complete renewal of the organism in each seven-year cycle, must have completely forgotten the nerve cell.

But one must not think that the life span of an animal depends only on the ability of its cellular elements to multiply and regenerate. On the contrary, observations show that the life span of various animals is also determined by a long series of internal and external factors. The flatworm, whose ability to regenerate excels that of man, does not live as long as man.

When a doctor pronounces a patient dead, not all parts of the organism are dead. The heart has stopped beating; the lungs no longer breathe; the nervous system has lost its excitability; the organism as a whole has

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ceased to exist. But many cells still retain life. A slight growth of beard appears on a dead man's face. The Russian professor, Kuliabko, succeeded in making a human heart beat, although it had been removed from the body more than 24 hours after death. The Moscow scientist, Briukhonenko, was able to bring dead dogs to life by artificially renewing their blood circulation.

Experiments in reviving a dead organism are difficult because as soon as circulation stops, the nervous system rapidly suffers great and irreversible biochemical changes. Yet one can keep the tissue cells alive for a long time in a specially prepared nutritive medium. Dr. Carrel, in his laboratory, put a piece of heart tissue, cut out of a chick, into an artificial nutritive medium and kept it alive many times longer than the average life span of a hen.

In the early period of the development of a fetus, a group of cells, without taking any part in body building, forms a special germinative tissue that eventually develops into male or female sex glands. These glands serve in the perpetuation of the species—in the endless array of changing generations.

From generation to generation this virtual transplanting of germinative tissue, called germ plasm, is repeated. In this manner descendants infrequently inherit the peculiar characteristics of a remote ancestor.

These examples of similarity with remote ancestors are not accidental. The hereditary characteristics are potential. Although they may remain hidden in a series of successive transplantings of germ plasm from parent to child, some hundred years later a particular characteristic may reappear in one of the descendants. This means that through the process of fertilization the germ

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plasm is transmitted from generation to generation, each time into a younger medium where it becomes rejuvenated, thus preserving its hereditary individual characteristics for many hundreds of years. When we speak of conserving hereditary traits in the evolutionary process of the species, however, the ability of survival of the germ plasm is measured in terms of many millions of years.

This germ plasm is nothing but cellular protoplasm. It illustrates the ability of living matter to prolong existence. The indispensable condition in the prolongation of life appears to be the periodical change of medium—the rejuvenation of the medium.

This condition is, of course, not the only one. The decisive factor is apparently the process of fertilization, the fusion of the male and female sex cells, that awakens the colossal creative powers in the produced embryo. However, the rejuvenating of the natural medium, as well as renewing the medium for the artificial tissue cultures, plays an important part in sustaining the longevity of the cells. This concept, it seems to me, opens before the medical profession new and wide perspectives in the struggle for longevity.

Medicine is faced with a problem of tremendous importance—to learn to control that inner medium in which the cellular elements live, to find methods for its systematic sanitation, cleansing, and renewal. It seems to me that contemporary medicine has already blazed a few trails toward the solving of this problem, whose significance man cannot overestimate.

In the further development of this thesis, the reader will become acquainted with some facts that concern this new direction in the fight for longevity.

5. *The Physicochemical Theory of Aging*

- 1 ALONG with the ability of cells to replace dying cells with new ones by multiplication, there is another phenomenon which is of great importance in the struggle against the wearing out of the organism. This is the ability of cellular matter to rebuild its own little particles, the so-called protoplasmic micelles,¹ which also
- 2 constantly change and disintegrate during life.

The structure of the cellular matter may be roughly visualized as consisting of very small particles, mostly albumin, suspended in a salt solution. These particles interact among themselves biochemically and physico-chemically. Only a few of these processes and the forces responsible for them are known to contemporary science. They are based on a series of physical phenomena such as variations in the electric charge at the surface of the micelles, viscosity of the suspension, and surface tension. To this may be added a series of chemical reactions effected by various ferments resembling those found in digestive juices, that cause the breakdown of foodstuffs and their subsequent synthesis into the substance of the living protoplasm.

The energy of the biochemical process in the cells depends to a large extent on the size of the protoplasmic colloidal particles—protoplasmic micelles.¹ The smaller they are the more easily they remain in suspension. By

¹ Micelle—a minute, colloidal unit of structure within the cell.

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the same token their motility is increased, thus making them more active biophysically and biochemically.

Have you ever noticed how ink "grows old?" Fresh ink has a uniform color, due to tiny particles of pigment suspended in water solution. But from long exposure to light, the ink begins to turn brown. On the bottom of the bottle appears sediment consisting of fine particles. The sediment gradually increases. Large brown flakes begin to settle to the bottom. The ink now is lighter but muddy. It no longer sticks to the pen point, and does not leave a distinct line on the paper. The ink has "aged," decomposed, lost its characteristics; in fact it has stopped being ink. The tiniest particles of pigment, that can not be seen before the ink begins to spoil, unite into larger particles too big to remain in the state of suspension in water. These large particles settle to the bottom and form a sediment. No matter how much the bottle is shaken now, this mixture will never be the same ink again.

This process of the aging of a colloidal mass (ink in this case) is very complicated from a physicochemical point of view. Even more complicated is the aging process of a living colloidal mass of which the cytoplasm (cell substance exclusive of the nucleus) is composed. The aging of cytoplasm begins as a result of the cell's loss of ability to destroy its dead albuminous micelles and to replace them with new, biologically active ones—a process which is so easily accomplished by a young cell possessing the ability for biochemical regeneration of its worn out parts.

Contemporary science understands the process of aging as a gradual weakening of cellular reactions. It

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begins with biophysical and biochemical changes in the cellular matter—the changing of its physicochemical structure. There is a gradual loss of the cell's ability to multiply and to renew its biochemical structural elements, and a clogging of the cell by the enlarged granules, which develop from its own cytoplasm.

It is accepted that the gradual loss of regenerative ability by cytoplasm is the cause of aging of the cell's colloids and micelles and of subsequent senescence and dying of the organism. This aging of the biocolloids coincides with a series of changes in colloidal properties and with a weakening of those physical forces in the living cells, that are responsible for physiological reactions. Under a microscope these changes are marked by a thickening and enlargement of particles in the cytoplasm and by the appearance in the cell of pigment granules and other biologically inert structures—the products of bio- and physicochemical degeneration of protoplasm. From the physicochemical standpoint the cells' aging is a matter of the thickening and condensation of cellular colloids, the decrease in their dispersal, the sluggishness of the protoplasm, the senile drying of the colloids, and the lowering of their physicochemical and biological activity. The nutrition of the cells is disturbed. Their starvation sets in. Life activity is lowered. Age comes on, then death.

The aging organism, as is often said, dries up. Wrinkles appear. The skin becomes thin, flabby and dry. This observation agrees entirely with what is known of physicochemical differences between an old and young organism. The water content of a month old fetus is about 97%. That of a baby is 80%. A grown

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man averages 64.5%. Aging is characterized by a further decrease in the water content of the organism.

It is a mistake to think that it is easy to help the organism overcome this loss of water. It is true that a young organism can easily replace a lack of water due to insufficient intake or excessive sweating. But the old organism dries up not because of an insufficient intake of water, but because the aged tissues of the muscles, skin, and internal organs have lost their ability to retain water. I want to emphasize again that the loss of water in the old organism is the result and not the cause of aging.

6. *Biologic Theories of Aging*

THE process of aging is extremely complicated. One of its usual manifestations, as has been pointed out, is an increase in the size of the particles in the cytoplasm and a decrease in the number of these particles. Because of this, the total surface of these particles, which provide the floor space for transacting the most complicated biological processes inside the cells, decreases sharply. This is strongly reflected in the quantitative and qualitative aspects of these processes, which slow down, and lose their normal intensity and lability.

The diminishing ability of the cells for biochemical regeneration has tremendous import in the struggle against the aging of the organism. Especially great is this importance for cells like those of the central nervous system that are incapable of multiplication and that maintain their life entirely through biochemical regeneration. With age some of these cells are destroyed and never replaced. It was estimated, for example, that in the cerebellum of a 14-year-old dog only $\frac{2}{3}$ of the original nerve cells were left and a 17-year-old dog had only $\frac{1}{3}$ of the number normally found in the brain of a young animal.

In the nerve cells, as in the rest of the cells of the organism, however, there is a continuous process of partial destruction and replacement—a process of rebuilding of the cytoplasm, a process of the cell's biochemical regeneration. Consequently, although the

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nerve cells do not reproduce—on the contrary, they are gradually destroyed—the nervous system in cases of exceptional longevity may retain its function for 150 years or more.

No matter how great the regenerative ability of the majority of cells in the organism, this ability is not limitless. With the passing years, it lessens and the organism becomes old.

What then is old age, "Old age that every man wishes to reach and upon reaching complains of . . . such is the inconsistency and foolishness of irrationality"?¹

In life's hustle we do not notice the swift flight of time passing by. Suddenly at a touch of the wings of time, silver threads begin to appear in our hair.

Old age creeps upon a healthy man in stealthy manner.

"He readily agrees that he looks ten years younger than his actual age—the age of his contemporaries. And then, suddenly, he begins to suspect the advent of old age. This may be caused by an accidental remark he overhears about himself, by an unusual expression on his face observed in the mirror, or by a charming girl, much to his surprise, offering him her seat in a crowded bus. When he takes a stroll, he notices that he is not as tireless as he used to be, but has to stop to catch his breath or to ease a pain caused by movement." (Rolleston.)

¹ "Cicero's Opinions", compiled by Abbot Olivet, translated by Captain Ivan Shishkin, published by the Russian Academy of Sciences, 1752, p. 87.

7. Metchnikoff's Theory

THE great Russian scientist Metchnikoff, who did much work on the problems of combatting old age, wrote: "With age men, as well as animals, undergo substantial changes. Strength fails, the body sags, hair grays, the teeth wear out. Briefly the signs of senile atrophy¹ begin to appear.

"In old age, which begins at different periods in different animals, the organism possesses little resistance to harmful influences and perishes from various pathogenic causes."

In his book, *The Nature of Man*, Metchnikoff expressed his views regarding old age as follows:

"Without a doubt it is erroneous to consider old age as a physiological phenomenon. One should not accept old age as the normal phenomenon, just because everybody ages, anymore than one accepts labor pains as normal. . . . One can use anesthesia for labor pains, but against old age—a chronic calamity—it is much harder to find a remedy.

¹ Atrophy means lack of nutrition, decline in nutrition. Atrophy is characterized by a decrease in the size of the cells, and by a weakening of their functions. One may consider senile atrophy as atrophy due to starvation. But the starving of the cells occurs in this case not because of lack of food, but as a result of inability on the part of the cells to assimilate food, to form new micelles to replace the worn out old ones. Hence it is concerned with the lessening ability of the cells for biochemical regeneration, of which we have already spoken. (A.B.)

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"The famous botanist Naegeli made the statement that there is no natural death in nature. He mentions trees that lived for several thousand years and met their end not through a natural death caused by exhaustion of vitality but because of some accident. It is supposed that the famous dragon tree of the Orotava Villa on Teneriffe that so charmed Humboldt, was several thousand years old. Its trunk was hollow, but the gigantic tree continued living until felled by a storm. So it took a rude outside force to kill this long-lived organism. It is commonly asserted that the baobab (an African tree) lives to an age of five to six thousand years."

Academician Liubimenko gives the following figures on the length of life of various trees: pine, 570 years; Siberian cedar, 700 years; beech, 1000 years; fir, 1200 years; Lybian cedar, chestnut, oak, 2000 years; cypress and yew, 3000 years.

Dracaena has been described in the literature as having reached the age of 8000 years. Tourists are acquainted with the olive tree of Plato, near Athens, that is more than 2000 years old. It is estimated that the platan of Hippocrates on the Island of Cos is more than 2300 years old.

In the forests near Kiev there are still many mighty oaks that lived during the epoch of Khmel'nitsky and possibly during the reign of Yaroslavl the Wise.

Of course, neither these oaks nor baobabs give us reason to consider them immortal. But a few thousand years is not too bad for the existence of a living organism, even if it is a plant. It is possible that the secret of this longevity, when discovered by science, may prove useful in the fight for the prolongation of human life.

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But to go back to Metchnikoff's teachings regarding the causes of old age:

"It is well known that the meat of old animals is tough. The tender meat of young chicks is not to be compared with that of old hens. The other organs, such as the liver and kidneys of old animals are much tougher than those of young animals. The tough meat of these old animals is often compared to shoe leather. This comparison is essentially correct. Shoe leather is made from animal hides; that is, from very tough tissue consisting of the so-called connective tissue. The connective tissue is composed of an enormous number of fibers mixed with the living elements, the connective tissue of cells. This tissue is very durable and for that reason is used for the manufacturing of shoes. . . .

"This hardening of the tissues is called sclerosis. Particularly in old age many organs of the body tend to harden or to become sclerotic.

"I formed my opinion as follows: In atrophy we always find the same picture—atrophy of the specialized tissues and their replacement by hypertrophied (enlarged in size—opposite of atrophy) connective tissue.¹ In the brain the nerve cells disappear and are replaced by neuroglia, a type of connective tissue. In the liver, connective tissue crowds out those cells that play an important part in the nutrition of the organism. This same type of tissue overruns the kidneys.

"In other words, old age is characterized by the struggle between the specialized elements of the organism and the common, primitive tissues, ending in favor

¹ Connective tissue—tissues that serve to bind together other tissues and to support the body.

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of the latter. Their victory results in the weakening of mental ability, disturbances in nutrition, and complications in metabolism, etc.

“In all parts of the body one may find cells that have retained a large independence. These cells can move about and are capable of ingesting various solid particles. For this reason they are called phagocytes or ‘devouring’ cells. Phagocytes perform an essential role in our organism. They gather in large numbers about microbes and devour them. Phagocytes also cause the disappearance of blood clots.

“In degeneracy due to old age, the phagocytes are responsible for the atrophy of the kidneys. They come to these organs in large numbers where they gather around the renal tubules and destroy them. Having taken their place, they form a connective tissue, which thus replaces the kidney tissue. An analogous process takes place in other tissues. It has been observed that in the brain of old men and animals many nerve cells are surrounded by macrophages (phagocytes) and are destroyed by them.”

The graying of the hair, one of the early signs of aging, according to Metchnikoff, is also caused by phagocytes that devour the hair pigment. “The mechanism of graying hair and fur indicates an increase in activity of macrophages, a phenomenon characteristic of old age. The porosity of the bones in the old depends on a similar mechanism; e.g. the nibbling and destruction of the skeleton by active macrophages invading the bone tissue.”

8. *Aging and the Reticulo-Endothelial System*

METCHNIKOFF'S discovery of the part played by the phagocytes' removal of dead matter from the organism is of enormous importance. This discovery explained the nature of the inflammatory process as a biological reaction, the basis of immunity to disease and the mechanism for developing such immunity. It shed light on the process by which wounds heal, on the mechanism for removal of dying parts from the organism. However, Metchnikoff committed a serious error. The "aroused phagocytes", he speaks of, are the result and not the cause of death of old cells. The phagocytes devour the dying, aging, lifeless cells, fulfilling, in this case, the function of scavengers.

However, Metchnikoff's basic error in presenting the picture of senescence consisted not so much in his describing the phagocytes as some sort of cut-throats as his failure to evaluate the importance for the organism of a healthy reticulo-endothelial system.¹

¹The term "reticulo-endothelial system" is used in this translation for what Bogomolets calls "the physiological system of the connective tissue." This is done because the functions he ascribes to the tissues in question are those that in American text books are ascribed to the reticulo-endothelial cells. This reticulo-endothelium according to our texts comprises the fixed histiocytes residing in or on (1) connective tissue; (2) reticulum of spleen, lymph nodes and bone marrow; (3) endothelial linings of the sinuses of liver, spleen, bone marrow, adrenal

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This nonspecialized tissue that Metchnikoff called base or primitive, exerts a tremendous influence on the specialized epithelial cells of internal organs and on the nerve cells. To understand the importance of connective tissue, we must review briefly the general plan of the structure of human organs.

Each organ—liver, kidney, brain—receives its food from the blood and returns waste material to the blood stream. The waste materials in the form of urine, carbon dioxide, etc. are expelled from the blood stream through the kidneys and lungs. But the specialized cells of these organs do not come into direct contact with the blood. The smallest arteries branch inside the organs into a system of very small, hairlike capillaries, which form a fine network around the cells. The specialized cells are located in the meshes of this net, adhering closely to the walls of the capillaries.

The capillaries, in their turn, form veins through which the blood returns to the heart, and through the right ventricle goes to the lungs, where it becomes saturated with oxygen and loses some of its carbon dioxide. After this, blood enters the left side of the

and pituitary glands; and (4) microglia; plus the wandering histiocytes in tissue spaces and the blood. "The physiologic system of connective tissue" of Bogomolets includes also the fibroblasts that make up the connective tissue itself and the specialized capillary endothelial cells of the entire circulatory system. It is, of course, generally conceded that fixed and wandering histiocytes reversibly change from one form to another, and that histiocytes may settle down to become fibroblasts or true endothelial cells.

To preserve the author's way of thinking, "connective tissue" and "reticulo-endothelial system" will be used synonymously throughout the book. *Trans.*

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heart from which it is pumped through the arteries to all parts of the body. In the kidneys the blood gets rid of other waste products, which are excreted as urine.

The blood passing through the intestinal walls becomes loaded with the products of digestion and carries them to the liver, where they undergo further changes.

The walls of the capillaries are very thin. They consist of a single layer of cells, called endothelium, which represents one of the varieties of connective tissue. The exchange between the blood and the specialized cells takes place through this wall. The nutrients pass from the blood to the cells and the waste products from the cells to the blood. Although the thickness of this membrane is only about $1/1000$ of a millimeter, its biological and physical condition affects greatly the state of the specialized cells. The slightest disturbance in the permeability of the capillary wall is sufficient to interrupt the supply of nutrients and the removal of waste matter.

Contemporary medicine pays too little attention to the effect of connective tissue, as a physiologic system, on longevity. The majority of learned medical men regard the connective tissue as simply supporting tissue, which acts as an elastic addition to the skeleton.

The term "physiologic system of the connective tissue" is still absent from medical manuals. They forget for instance that a disturbance in permeability of the capillary wall alone, which takes place during sclerosis of senescence, leads to cell starvation and autointoxication.

The connective tissue in the organism plays not only

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the role of a supporting tissue and membrane that separates the specialized cells from the blood, but, as Metchnikoff pointed out, it takes an active part in the reaction of the organism to infection. The course of such diseases as syphilis and tuberculosis is greatly determined by the ability of the cells of the connective tissue, as a physiologic system, to withstand infection and destroy microbes.

One of the greatest achievements in medicine is the diphtheria antitoxin. This discovery and its introduction into medical practice eliminated great diphtheria epidemics, which used to have a death rate of 40 per cent. This antitoxin is present in the serum obtained from the blood of horses that have been inoculated with the diphtheria toxin. The cells of the connective tissue absorb this diphtheria toxin, develop antitoxin, and discharge it into the blood stream. In a similar manner are produced antisera against other infectious diseases.

Connective tissue cells react intensively to stimuli. The healing of wounds is an illustration of this reaction. The cells grow toward each other, reinforcing the edges of a wound with their fibers. These fibers gradually solidify and form a scar.

The vigorous reactivity of connective tissue plays also an important part in the development of cancer. Connective tissue responds by an abundant growth around the tumor, penetrates into the tumor itself and tends to prevent its spread into the surrounding tissues. The connective tissue cells destroy the tumor by means of phagocytosis and by dissolving it with the aid of their enzymes.

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Using an anti-reticular cytotoxic serum,¹ introduced by me, Dr. Neiman and I succeeded in strengthening the reaction of connective tissue to carcinoma, thus raising the resistance of mice to that disease and in many cases causing the disappearance of large tumors. These experiments are now being tried on human beings, by Professor Kavetsky and his assistants in the Institute which I direct. They have demonstrated, however, that the serum suggested by me does not cure cancer in man. The tumor must be removed by a surgeon. But it is hoped that, with the help of our serum, we may succeed in preventing the return of cancerous growth, now so frequent after surgical removal of a tumor. The tiniest particles of a tumor, that often escape the surgeon's knife and grow large again, may be destroyed by the connective tissue cells that have been stimulated by our serum.

When one considers that the death of 10 to 15 per cent of people who die after reaching 50 years of age is caused by cancer, and that cancer frequently recurs after an operation, it will become clear how very important, in the fight for longevity, is the possibility of increasing the resistance of an organism against malignant tumors. I stated a long time ago that cancer cannot develop in an organism if the reticulo-endothelial tissue has retained a sufficient degree of resistance. This point of view is gaining more and more followers.

The connective tissue of the organism plays the part of a depot for nutritive substances and, at the same time, acts as a regulator in the processes of metabolism.

¹ Anti-reticular cytotoxic serum—A.C.S., a serum that has a specific effect on the reticulo-endothelial system.

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The reticulo-endothelial tissue may be compared to the root of the organism. As a plant receives its nourishment from the soil through its roots, so do the specialized cells of the animal receive all they need from the blood stream via the connective tissue. And as the plant's growth, development, and general appearance depend on the state of its root system, so does a man's constitution depend upon the condition of his reticulo-endothelial system.

The biochemical characteristics and microscopic structure of connective tissue enable us to distinguish four basic human constitutional types: namely, 1) lipoidal (fatty), 2) lymphatic (pasty, puffy), 3) asthenic (with thin, tender connective tissue), and 4) fibrous (with solid, strong, connective tissue). The kind of disease to which an individual is predisposed and the chances for his longevity depend in a large measure on the type of constitution he possesses.

French doctors have long said that a man is as old as his arteries. This is true. Early arteriosclerosis, which, for example, is often a result of syphilis and alcoholism, causes the heart to wear out, because of the loss of elasticity of the arteries. However, arteriosclerosis is only one sign of the many common scleroses of connective tissue. The saying of the French doctors could be paraphrased: Man is as old as his connective tissue. This leads us to the conclusion that the fight for longevity should be, to a great extent, the fight for healthy connective tissue.

Metchnikoff's failure to evaluate the role of connective tissue (the reticulo-endothelial system—Transl.) was the basic error in his theory of senescence. His other

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error, already pointed out by many scientists, was in not recognizing that changes in the connective tissue of an aging organism come either at the same time or earlier than changes in the specialized cells of the nerve system, liver, kidneys, and other organs. The latter changes, in my opinion, result largely from alterations in the connective tissue. And only after the specialized cells have undergone these changes do they become the prey for phagocytes.

As the connective tissue loses its physiologic elasticity, the body's functions gradually lose their flexibility, adaptability and capacity. The mental power is still at its height; the digestive juices still flow normally; the kidneys produce the same concentrated urine. One still does not suffer from shortness of breath or dizziness, but already the tale-telling wrinkles begin to form around the eyes and the deep sad lines appear around the mouth: life's parabola has passed its zenith. Old age is coming on.

9 *Endocrine System and Aging*

IN THE bodies of higher animals certain small organs are indispensable to growth and development. The physiological functions of these organs have been explained only in the last few decades. These organs secrete their products directly into the blood stream and are, therefore, called the glands of internal secretion or the endocrines. Although these organs are located anatomically far apart, contemporary physiology has unified them into a single system of internal secretion—the endocrine system. The reason for this is that although the products of these glands differ widely in their chemical composition, as well as in their physiologic action, they have one fundamental similarity. They are secreted in small quantities into the blood stream and thus are carried throughout all organs and tissues. They regulate metabolism and stimulate specific functions of the cells. For this reason they all have one common name—hormone, which in Greek means a stimulator, an exciter.

I shall mention the most important of these glands.

At the base of the brain is found one of them called the pituitary gland (hypophysis). It is the size of a pea, but it very complicated both as to structure and function. The pituitary produces a substance that stimulates and regulates the growth and development of the human organism. If, for example, the pituitary gland be removed from a puppy, this animal will grow very

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little, get fat, never reach sexual maturity, never completely develop physically and mentally, and will age and die long before the end of its normal life span.

In the front of the throat, straddling the windpipe, lies the thyroid gland, which produces a substance containing iodine. This influences growth and development of the body. Suppression of the functions of the thyroid gland at an early age, results in cretinism and premature aging.

The pancreas, which secretes directly into the intestinal canal a juice rich in enzymes¹ and essential for digestion, produces also a substance called insulin, which goes directly into the blood. The word insulin comes from the Latin word *insula*—island, because this secretion is produced by special cells that are scattered throughout the body of the pancreas like little islands. Insulin regulates the utilization of sugar by the organism.

Close to the kidneys are located two small glands—the adrenals. Their inner portions produce adrenaline, a hormone regulating the function of the sympathetic nervous system which in its turn controls, to a great extent, the functions of the body. The external layer, cortex of the adrenal gland, secretes the hormone cortin or the hormone of relaxation. My investigations and especially those of Professor Nina B. Medvedeva have shown that this hormone counteracts an autointoxication caused by muscular work. It also regulates carbohydrate metabolism and the formation of the esters of phosphoric acid (hexose and creatine), substances of great

¹ Enzymes or ferments or catalysts—biochemical agents that produce specific changes in the physical and chemical form of substances, present in all body cells, sometimes secreted as in digestive juices.

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dynamogenic importance. An animal deprived of both adrenals dies within a few hours after the operation.

By means of their hormones, secreted into the blood stream, the endocrine glands affect each other and the rest of the body, thus bringing about a chemical regulation and co-ordination of its functions. A disturbance in the function of the glands of internal secretion causes a disturbance in the growth and development of the organism, its untimely wasting away, and sometimes a fatal autointoxication. It is no wonder, therefore, that contemporary biology is inclined to ascribe an important role to the exhaustion of the endocrine system in the aging of the organism.

A special place in the study of longevity is occupied by the problem of the internal secretion of the sex glands. Long before the discovery of the functions of the endocrine organs, scientists made conjectures about them. For example, at the end of the 18th century Hufeland in his book, *Makrobiotik*¹ (study of longevity), wrote: "The male semen is intended not only for excretion, but even more so for absorption by the blood for our own invigoration." The semen does not go into the blood stream. But the male and female sex glands secrete into the blood, substances that greatly influence the organism.

Of course, everybody is greatly impressed by the changes that take place in the organism during the period of sexual maturity. They are, to a large extent, caused by substances that are produced and secreted into the blood stream by matured male and female

¹ Hufeland, C. W. *Makrobiotik, oder die Kunst, das menschliche Leben zu verlängern*. 1796; published in Russian, Moscow, 1852.—Transl.

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sex glands. These substances are responsible for sex differentiation and for the development of psychological and physical secondary sex characteristics. Early castration prevents the organism's maturing, thus leaving it in an "intersexual" stage and causing external signs of premature aging.

Experimental physiology has succeeded in transplanting sex glands of males into females and vice versa, causing effeminacy in male and masculinity in female organisms, even to the extent of psychological manifestations. A capon develops the feathers and instincts of a hen. In a feminized male guinea pig mammary glands enlarge and may produce milk.

Just think of the complicated evolution of the female organism; first an infant, then a girl, then a mother, and then an old woman. All these periods, from infancy to menopause, so sharply different physiologically and psychologically, are the result of actions of the hormones produced by the sex glands—substances of a relatively simple chemical composition.

We still know neither the nature nor the mechanism of these changes, which arise as a result of specialized biologic reactions of the cells to substances of a definite chemical composition. Many of the hormones have already been synthesized by chemists. A striking observation was made that substances which are very closely related chemically to sex hormones may cause experimental cancer. This led to the supposition that cancer sometimes may be the result of a vicious change in the chemical nature of the hormones secreted by the aging sex glands. Experimental medicine is full of questions of tremendous academic and practical interest.

10. *Nervous System and Aging*

THE condition of the nervous system is very important for the well-being of the organism, for its responsiveness, and for its longevity. Particularly important is that part of the nervous system which regulates the functions of the internal organs: the rhythm and force of heart contractions; the tone of blood vessels; the secretion of digestive juices, urine, and hormones; the biochemical processes of assimilation and dissimilation in the liver and in other tissues and organs. This part of the nervous system is called vegetative or autonomic, since its activity is not subject to conscious control or will as is that of the motor nerves that govern the skeletal muscles. We cannot at will hasten or slow down the secretion of gastric juice, the beating of the heart, or the movement of the alimentary canal, because these are regulated by the vegetative nervous system. However, the vegetative nervous system is only relatively independent of the centers of psychic life. Between them there exists a very close physiologic connection. Particularly does this remark apply to that part of the vegetative nervous system which is called sympathetic.

Its very name, "sympathetic" nervous system, implies a connection between it and life's psychic manifestations. Study of the function of the sympathetic nervous system more and more confirms the point of view that its role is to sustain the physiologic intensity, the tonus of the functioning cell. Tonus, i.e. the intensity

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of the physiologic processes as well as the tonus of the organism as a whole, is regulated to a large extent by the sympathetic nervous system.

The degree of intensity of reactions of the organism, their co-ordination, and their proper relation to stimuli depend largely upon the tonus of the sympathetic system. These are further affected by the activity of the peripheral organs, which, although being themselves regulated by the sympathetic system, exert a reciprocal effect upon it. This particularly applies to the glands of internal secretion, whose products have such a strong influence on the sympathetic nervous system that many physiologists are inclined to regard the endocrine and vegetative nervous systems as a single neuro-endocrine system.

Although the general condition of the organism and the state of the internal organs are not subject to conscious control, there is no doubt that they are affected by higher nervous activity—psychic activity, through the medium of the sympathetic nervous system.

On the other hand, there is no question about the influence of the sympathetic nervous system on the manifestations of the psychic life. The evidence is supplied by the fact that the nutrition of the centers of the psychic life, their blood supply, and the intensity of the biochemical processes that go on within the nerve cells are regulated by the sympathetic nervous system.

Physiological psychology has long evaluated this interdependence by making the well-known, although paradoxical statement: "We are sad because we cry."

Feelings of joy, fear, and anger accelerate the heart beat, cause blushing and paling, all as a result of

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changes effected in the functioning of the sympathetic nerves. The following examples illustrate the connection between the central and vegetative nervous systems: cold sweat caused by fear or pain; peristaltic disturbances of the intestines resulting from the same causes; and emotional glycosuria (sugar in the urine). The flow of saliva and other digestive juices at the thought of food and sexual excitation caused by erotic imagination belong in the same category.

Medicine has accepted the idea of a powerful emotion called shock. It is really possible to "die of happiness," or of grief, or from a sudden strong emotional upset. A French writer cites many such examples. Sophocles died during the applause of the crowd that crowned him as their genius. Diagoras died suddenly when he learned that his three sons were victorious in the Olympic games. The niece of the philosopher Leibnitz died of happiness when she found six hundred thousand francs hidden in her uncle's deathbed. Pitt died when he learned of Napoleon's victory at Austerlitz, etc.

Here is what Hufeland wrote in his book, *Macrobiotia*,

"Among the influences that tend to shorten man's life, a foremost place is occupied by moods and emotional habits: sadness, depression, fear, longing, timidity, envy, and hatred.

"Fear is a shameful emotion that destroys a man's dignity. Fear deprives man of his strength, his perspicacity, his reason, determination—all the human qualities he possesses.

"Fear is a continuous spasm. It contracts all the capil-

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laries. The entire skin pales and grows cold. All the blood collects in the internal large blood vessels. Pulsation of the arteries stops. The heart overfills and cannot beat freely. Digestion is greatly disturbed and cramps and diarrhea occur. The muscles become paralyzed. The man wants to run and cannot. He trembles all over. Breathing is short and irregular. Thus fear brings about all the symptoms that are typical of a slow-acting deadly poison and, consequently, shortens life. But no other fear makes man more miserable than fear of death. Whoever is afraid to die will find no happiness in life. Schiller once said: 'He who has faced death is a man freed.'

"Sadness, worry and fear help to bring on old age. There are instances of people who, subjected to fear, turned completely gray in the course of one night."

Alongside the fear of death there is also mortal boredom. In this connection Hufeland wrote:

"Boredom is very dangerous both physically and mentally. What do we notice about a man who is bored? He begins to yawn. This shows that the blood's passage through the lungs is encumbered. Evidently the strength of the heart and of the blood vessels is impaired. Local congestion of the blood takes place. Digestive organs are weakened and flabby. Fatigue, heaviness, and indigestion appear. The man becomes a hypochondriac.

"Not a single lazy man ever reached old age. Old age is reached only by those who lead a busy life.

"It is impossible to reach old age with a bad stomach. There is a great deal of truth in the statement that a good stomach is not affected by aggravations, sadness, or insults. All these emotions affect primarily the stomach.

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"Of all man's functions that affect the body and soul together, laughter is the healthiest. Laughter aids digestion, circulation, sweating, and has a refreshing effect on the strength of all the organs."

We find analogous conclusions among the medics of ancient times, beginning with Hippocrates.

In his book *Bodily Changes in Pain, Hunger, Fear, and Rage*, the well-known contemporary physiologist Walter Cannon arrived at the same conclusions regarding the strong influence that the psychic life exerts on the circulation, on digestive and endocrine secretions, and on the general nutrition of the organism.

We have talked at length of the importance of the condition of the sympathetic system in sustaining longevity. We also talked about the usually forgotten fact that the sympathetic system in its turn is affected by all psychological experiences. I would like to remind you that angina pectoris, so common after the age of fifty, begins in many cases with, or is aggravated by, disturbances in the function of the vegetative nervous system. One must control one's self. Unusual irritability, which leads to quarrels, shortens life.

II. *Chronic Poisoning of the Organism*

WE MUST also consider chronic poisoning of the organism as a cause of untimely aging. The organism is systematically poisoned throughout its entire life. The poisons come from the outside world in spoiled food, in polluted air, or in improper diet.

The Italian, Luigi Cornaro, a man of weak physique and poor in health, wrote in 1558 a book entitled, *The Sure and Certain Method of Attaining a Long and Healthful Life*. Cornaro lived to an age of over 100 years. He attributed his long life and his always excellent disposition to moderation in everything, particularly in food and drink. He ate only 360 grams of solid food and drank about 400 grams of liquid every 24 hours. At the age of 83 he still rode horseback, climbed hills, and wrote a gay vaudeville skit.

Cornaro became the fountain head of a new movement—the fight for longevity through partial starvation and through reduction of metabolism to a minimum. Followers of this movement are to be found even now. I do not belong to their number. A starvation diet leads to the weakening of the organism. It can, however, be very useful as a temporary cure for certain diseases of metabolism. When prescribing such a diet, the doctor should consider it not only quantitatively but also qualitatively, so as not to deprive the organism of essential materials for the rebuilding of cells and for the avoidance of some forms of avitaminosis.

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To say nothing of acute serious infections, such as typhus, diphtheria, scarlet fever, which sometimes leave a lifelong mark on such important organs as the heart and the kidneys, great harm is done to the organism by the so-called minor ailments: grippe, sore throats and colds. Tuberculosis and syphilis, because they spread so rapidly and cause serious disturbances in the functions of the organism, are particularly important obstacles to attaining a normal, long life. Both these diseases are particularly harmful to the reticulo-endothelial system.

Syphilis is the most frequent cause of sclerosis of the heart and blood vessels, and of angina pectoris. It also causes sclerosis of connective tissue which, as we have said, is so important for nutrition and activity of the human organism. Scars, that remain in internal organs after the healing of syphilitic lesions, often become the seat of cancer. The sclerotic connective tissue, which forms scars, contains no cells capable of resistance to the development of cancer. A most active crusade against the spread of the above mentioned diseases should be undertaken jointly by both the individual and society. This crusade should become an indispensable part in the common fight for normal longevity of man.

The saying that "milk is children's wine and wine is old men's milk" is not without some basis. A limited use of alcoholic beverages as a medicine for improving the appetite, for stimulating the secretion of digestive juices, may undoubtedly be useful. Too frequently, however, alcohol is used excessively by people who do not consider themselves as drunkards. Alcohol in large doses is the strongest poison for all cells of the body,

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especially for the nervous system and for the reticulo-endothelial tissue. Alcohol stimulates the proliferation of connective tissue and its subsequent thickening—sclerosis.

Whether nicotine causes sclerosis of blood vessels has not been completely proved. Nicotine is unquestionably harmful to the sympathetic nervous system and the blood vessels. A spasm of the arteries feeding the heart wall often is observed as the result of excessive smoking and produces the symptoms of angina pectoris. Although in medicine this angina is called "false," and its attacks stop as soon as excessive use of tobacco is discontinued, the heart muscle, nevertheless, may severely suffer for the rest of life from the interruptions of nutrition. The insufficient supply of blood to the heart muscle during the spasm causes tiny but numerous areas of atrophy in the muscle fibers, which are subsequently replaced by small scars made of connective tissue. Such heart muscle can never fully recover and consequently weakens prematurely.

Although nicotine is a strong poison, not everyone is equally sensitive to it. People who are predisposed to circulatory spasms (migraine, blanching of the fingers, and intermittent limping) must absolutely refrain from smoking.

Sometimes the spasms provoked by smoking are so persistent that, for example, the pulse in the artery passing over the instep of the foot may disappear and numbness develops in the toes.

By persuading a hunting companion of mine to stop smoking, I literally saved him from an amputation of the foot to which he was condemned by the surgeons.

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A month after he had stopped smoking, all the signs of incipient atrophy of the toes had disappeared.

I do not wish to frighten the reader. My aim is to help him reach his normal age; moreover, fear in such an undertaking is a poor help. Nevertheless, I must mention one more instance of chronic poisoning of the organism that Metchnikoff particularly urged us to fight against.

According to Metchnikoff, "sclerosis without a known cause is clearly not a physiologic phenomenon; it must be as much pathological as is sclerosis of syphilitic and alcoholic origin. In order to explain arteriosclerosis in patients who do not suffer from syphilis, alcoholism, or other diagnosed causes, the blame must fall on the innumerable microbes that swarm in our intestines and poison us."

How large is this bacteria population of the intestines? We can have an idea regarding this from the fact that the intestinal bacteria comprise about one-third of the feces. According to a count made by Strassburger, the body excretes in this manner about one hundred and twenty-eight trillion bacteria daily.

Metchnikoff said, "Bouchard called attention to the part played by the intestinal canal in the intoxication of the organism. He especially emphasized the role of the toxins that come from food, in the origin of arteriosclerosis. These toxins are manufactured by the intestinal microbes."

The poisons originate in the large intestine. They are produced from the waste products of food, whose valuable parts have already been digested in the small intestine and absorbed into the blood stream.

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According to Metchnikoff, the large intestines, where ptomaines that poison and age the organism are formed, should be removed. Said he: "Already many leading surgeons dare to remove the large intestines, when patients suffer from habitual constipation. However, for the present it is more sensible to deal directly with the microbes populating our intestines." A series of studies revealed that "the microbes that cause the souring of milk by changing lactose into lactic acid are known for their action against the ptomaine-producing bacteria.

"And so it is quite clear, that in order to stop this slow autointoxication, we must include kefir (a specially fermented milk, much used in the Caucasus), or still better, clabber in our diet. The presence of large numbers of lactic acid bacilli will interfere with the development of putrefactive bacteria. This alone will be very beneficial for the organism."

In the course of the biochemical processes that go on within the cells of a healthy organism, some substances are produced which possess poisonous properties. Most of these substances are made harmless by the liver and kidneys and are excreted with the urine. However, the urine of even perfectly healthy people is poisonous. Disturbances in urine excretion which occur during diseases of the kidneys cause a serious condition called uremia, (literally—urine in the blood) which frequently leads to death.

A partial deficiency of the kidneys may last for years. It causes a chronic poisoning of the organism which in turn causes premature aging and shortening of life.

A disturbance in the internal secretion of the pancreas, as we have already stated, causes diabetes. In

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diabetes, metabolism is affected and poisons are formed in the organism. Inadequate functioning of the heart or the lungs also causes autointoxication of the organism, because the disturbed oxidation and blood congestion in the organs leads to inadequate removal of the waste products.

Of particular importance in disturbances of metabolism is the irregular functioning of the glands of internal secretion. An excess or lack of certain hormones also leads to autointoxication of the organism and shortening of its life.

Excessive activity of the thyroid gland causes so-called Basedow's disease and a rapid wasting away of the organism as a result of the disruption of co-ordination between the processes of assimilation and dissimilation.

Insufficient activity of the thyroid gland causes a slowing of the metabolism which also leads to autointoxication, lowers resistance to infection, weakens mental abilities, and shortens life.

We have already mentioned that substances closely related chemically to sex hormones may cause cancer. Although as yet there is no proof that these substances form during a disturbance in the internal secretions of the sex glands, the possibility that they do form is likely. Other chemically related substances (derivatives of stearin), according to the opinions of many scientists, bring on arteriosclerosis when they accumulate in the blood stream. Disturbances of metabolism and of the glands of internal secretion lower the vitality of the reticulo-endothelial system, thereby causing its hardening and aging and also an untimely aging of the entire organism.

12. *Normal Longevity*

THE reader can see the great number of obstacles, both internal and external, that stand in the way of reaching a normal old age. He must have noticed however, that a great many of these obstacles are entirely removable, thanks to our present day scientific knowledge. Further advances in medical science will open wide the road to normal longevity.

But what is normal longevity? What is the normal life expectancy of the human organism? Is contemporary science able to determine the normal limit of longevity of an organism placed in ideal conditions from the moment of its conception?

Let us turn to facts and observations. They indicate that under natural conditions the different species of animals, birds and mammals, have different lengths of lives.

Hufeland, in the book mentioned before, wrote that in 1732 a falcon from the Cape of Good Hope arrived in London, bearing a gold marker with the inscription: "His Majesty's James I, King of England, 1610." The falcon lived 122 years in captivity and it is not known how long he had lived before that.

Hufeland went on to say:

"The crow lives to be one hundred. So does the swan. The peacock lives to the age of twenty, the rooster to a decidedly shorter age, because he is such a tempestuous, quarrelsome, and lustful fowl.

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“The life span of the sparrow, lady’s man among birds, is even shorter. The blackbird and goldfinch live the longest among the small birds—twenty years.

“The bull is large and strong, but does not live long—from fifteen to twenty years. On the other hand, the camel lives much longer. By nature it is a thin, spare animal, active and very strong. The camel usually reaches the age of fifty but often lives to one hundred.”

Aristotle pointed out that the longer it takes an organism to reach a full growth, the longer is its life span. This idea was shared by many outstanding naturalists and medical men. Buffon considered that the probable life span of an animal is five to seven times the length of the period required to reach a full growth.

If we assume that the growth of an organism ends when the bones stop growing in length, then the coefficient of longevity suggested by Buffon is more or less correct as far as domestic animals are concerned.

ANIMAL	DURATION (IN YEARS)	
	<i>Growth</i>	<i>Longevity</i>
Dog	2	10-12-15
Cat	1.5	8-10
Ox	4	20
Horse	5	20-30
Camel	8	40

The above figures show that actually the span of life exceeds the period of growth at least five times. Man completes his full growth between twenty and twenty-five; hence, according to Buffon’s coefficient, man’s average normal life expectancy should be one hundred and fifty years.

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I cannot name a single authority who has studied the problem of man's longevity and who has not asserted that man's death before he is one hundred years old is due to the accumulation of unfavorable circumstances such as illness, overwork, lack of personal or social hygiene. (In Russia, a century is synonymous with "one hundred years" and "a lifetime.")

Haller, the famous physiologist of the 18th century, considered the limit of man's life to be not less than two hundred years. Hufeland holds the same opinion: "Man's organism is capable of living and functioning to the age of two hundred. It is permissible to accept that an animal lives eight times longer than the time it takes to grow up. To achieve complete growth man needs twenty-five years. Therefore, his life span without a doubt may be considered as two hundred years."

According to Metchnikoff the limit of man's life should be higher than one hundred years.

But Buffon's coefficient of longevity is in reality an empirical observation, which cannot be considered a law of nature, although it is supported by the data obtained on domestic animals.

This coefficient cannot be fully applied to birds. One can hardly agree with Hufeland that "the sparrow's shorter life" is exclusively due to the fact that the sparrow is "a lady's man among the birds." Metchnikoff tried to show the relationship between the longevity of birds and the length of their large intestines, on which depends the degree of autointoxication by the products of intestinal putrefaction. Subsequent investigations, however, have not supported this idea.

We must admit that contemporary biology has not

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established objective scientific facts which can be used as a basis for determining the norms for longevity among various animals. Biology usually in these cases points to heredity, which transmits to the animal certain traits characteristic of a species, among them the normal span of life. But this, of course, is not the explanation of the essence of the processes occurring within the organism and conditioning the degree of its viability. Biology and medicine should try to discover this essence because only as we come to know it, may we learn, to an extent, how to govern it. This can be accomplished only through experimental research.

13. *Examples of Man's Longevity*

THE present knowledge of heredity is too empirical to be of much help in solving the problem of longevity. The only real indicators of the possible length of man's life are the actual records of the maximal longevity. World literature has already a very large number of these records. I shall cite some of them as given by Legrand, Hufeland, Rolleston, Lacassagne, and a few others.

Hufeland reported that "Kentigern (St. Mungo), the founder of the Episcopacy of Glasgow, lived to the age of one hundred and eighty-five. . . . In 1724 in Austria, P. Ktzarten died at the age of one hundred and eighty-five. His son was then ninety-five years of age. In 1670 in Yorkshire, G. Jenkins died at the age of one hundred and sixty-nine. The famous Thomas Parr lived for one hundred and fifty-two years the life of a hardworking farmer. At one hundred and twenty he married again—a widow with whom he lived for twelve years. He was so energetic that his wife did not notice his real age. Parr outlived nine English kings. He was invited to the Court and died there from overindulgence in food and drink. The famous Dr. Harvey performed an autopsy and found no changes in the organs due to age. In Norway, in 1797, Joseph Gurrington died at the age of one hundred and sixty. He left a young widow and many children from several marriages. His oldest son

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at that time was one hundred and three and his youngest was nine."

Rolleston tells of the Hungarian, John Roven, and his wife Sarah, who lived in connubial bliss for one hundred and forty-seven years. John died at the age of one hundred and seventy-two, and Sarah at one hundred and sixty-four years of age.

According to Legrand, a Dane, Drakenberg, lived to the age of one hundred and fifty-six years. When he was sixty-eight years old, he fell a prisoner to the Arabs and lived in slavery until he was eighty-three. At the age of ninety he still worked as a sailor. At the age of one hundred and eleven he married. When his wife died, he tried to remarry but was turned down. The painter Kramer painted a portrait of Drakenberg at the age of one hundred and thirty-nine. He then looked like a strong old man. He died in 1772 from drink.

Humboldt reported that he met a couple in Peru, the husband was one hundred and forty-three and his wife was one hundred and seventeen. Finot, the author of the book *The Philosophy of Longevity*, told that in Hungary in 1905 a one hundred and ninety-five year old peasant died. His son was one hundred and fifty-five years old.

Gueniot, a member of the French Academy of Medicine, author of the book *To Live One Hundred Years*, who himself lived one hundred and three years, warned against excessive gullibility when dealing with people supposed to be over one hundred years old. He pointed out that it is essential to verify the ages through documents. However, he also presented a series of cases of exceptional longevity the correctness of which

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leaves no doubt. Among them is the one hundred and fifty-two year old T. Parr, already known to the reader, and the even older G. Jenkins, who was born in 1501. When summoned to court, Jenkins stated under oath that he was one hundred and forty years old. He died in 1670 at the age of one hundred and sixty-nine years. Elizabeth Durieux, an innkeeper born in Lyons in 1717, died in 1857 at the age of one hundred and forty.

Gueniot also tells the following interesting episode from the remote past: On July 31, 1554, Cardinal d'Armagnac, passing on the street, saw an eighty year old man crying at the door of his own home. When questioned by the Cardinal, the old man answered that his father had given him a beating. Surprised, the Cardinal expressed a desire to see the father. An able-bodied man of one hundred and thirteen was presented to the Cardinal. The old man explained that he had beaten his son because he showed lack of respect for the grandfather, whom he had passed without a greeting. When the Cardinal entered the house, he saw a third old man—one hundred and forty-three years of age.

In 1809 Doctor P. Defournel, the author of the book *Nature Unmasked*, died at the age of one hundred and nineteen years. In 1792, at the age of one hundred and two, he married a twenty-six year old girl and had several children by her.

The reader might say that all this happened too long ago—the Glasgow Episcopacy, Parr, John Roven and his wife Sarah, and the other examples of exceptional old age. For this remark he deserves an admonition for unfounded skepticism. I desired to show by the above examples that even one hundred and fifty years is not

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the limit of man's life, and that in some cases man can live much longer, sometimes even in spite of unfavorable environment.

Let us turn now to contemporaries, to present day citizens of the great USSR. Unfortunately, I have not at my disposal the latest census figures, which, when they are tabulated, will undoubtedly uncover tens of thousands of people whose age exceeds one hundred years.

My own illustrative material has been gathered from various notices in the press, and it is too large to be fully used in this small book. It suffices to show that one hundred years, far from being the limit of human life, cannot even be considered as the limit of his full physiological activity and ability to work. Here are a few illustrations:

In 1927 Henri Barbusse visited the village of Laty near Sukhum, where peasant Shapkovsky lived. He was one hundred and forty years old. Barbusse was amazed at his vitality, the speed of his movements, the resonance of his voice, and the clearness of his eyes. His third wife was eighty-two years old. His youngest daughter was twenty-six. Thus beyond the age of one hundred and ten, Shapkovsky was still virile. In 1927, too, in Novoborisov (in White Russia) lived Martsiana Maliarevich. She was born in 1796. At the age of one hundred and thirty she walked some twelve miles for a registration.

Numerous instances of unusual longevity occur in the villages along the river Ulakh in the Far East. The oldest man, L. Postnoy, is now one hundred and thirty-six years old. He is still active, helps about the household,

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and chops wood. (Quoted from the newspaper *Viechernaya Moskva—Evening Moscow—1935.*)

"In Odessa lives G. Katsaf, reported to be one hundred and ten years old. His mother died in 1928 at the age of one hundred and thirty-seven. He is still active, reads without glasses and has worked until the age of one hundred and five." (*Izvestia*, 1935, No. 57.) "Recently he celebrated his 115th birthday. As ever he is full of life and happy." (*Soviet Ukraina*, 1939, 1-11.)

"The one hundred and thirty year old Esbeyevev lives in the village Karisut of the Karachayevsky Autonomous Region. He is well and still hunts." (*Pravda*, 1935, 4-27.)

"At the Clinic for Old Age Pathology (VIEM), recently one hundred and twenty-three year old Mashukhin died of an infectious disease. He was born in 1811. The year of his birth was confirmed in the village records. Mashukhin stopped farming only three years before he died. To the last days of his life Mashukhin possessed an excellent memory, had fairly good hearing and eyesight. He never drank or smoked. He left an eighty year old daughter." (*Izvestia*, 1935, No. 10.)

"The village soviet of Kosinovsk, Kursk Region, recently celebrated the 115th birthday of a woman, Mezéntseva. Her oldest son, who is ninety-two is an instructor on a collective farm." (*Magazine Ogoniok*, 1936, No. 4.)

"In the village Karpilovka, of the Ostersky District, lives a one hundred and thirty year old woman, Ouliana Yakimenko. Her husband died thirty years ago at the age of one hundred and one. Of Yakimenko's twelve

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children all reached ripe old age." (The newspaper *Communist*, 1936, VII, 4.)

"Members of a kolkhoz (collective farm) in Gali, Abkhazia, celebrated the 132nd birthday of Ketsba Tlabagan, the oldest inhabitant of this settlement. Tlabagan was the first settler of Gali and the living witness of many historical events of the 19th Century."

Under the heading "The Land of the Oldest People," the newspaper *Komsomolskaya Pravda* (*Young Communist Truth*), 1936, No. 42, reported many instances of outstanding longevity among inhabitants of the Ochetchirsky District, in Abkhazia. "Only a year ago Khapara Knut died at the age of one hundred and fifty-five years. He rightfully could be considered the oldest man in the world. Now Adleyba Madczachva claims this title. He lives in the village of Tchilov. Madczachva is one hundred and fifty years old. He looks well; has an excellent memory. He is an inveterate reader of the local paper, published in his native tongue."

It was reported from Grozny that in the village Elistandzi, in the Vedensky District, lived one hundred and sixty-six year old Gunakbay Geziev and one hundred and forty-seven year old Khansimurad Dadayev. Both of them knew personally the popular hero, Shamyl. (1798-1871, leader of the Caucasian Rebellions—Transl.)

Vasily Nikiforovich Ivanov of the town of Minusinsk has just passed the age of one hundred and thirty-four. He feels fine, hears and sees quite well. Two of Ivanov's great-grandsons were called into the Red Army in 1939. (*Pravda*, September 25, 1939.)

In the USSR it is possible not only to live to the age

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of one hundred and one hundred and fifty years, but also to retain the honorable status of an active member of society. Under the heading, "A Noted Jubilee," the newspaper *Soviet Ukraine* of December 3, 1939, carried the following communication from Odessa: "The members of the collective farm, Shevchenko of the Krivozersky District, are celebrating a remarkable jubilee. The well-known shepherd, Tovaristch Kozorez, just passed the age of one hundred. Tovaristch Kozorez is one of the best shepherds in the Ukraine. Despite his age, he achieved remarkable success in the realm of cattle breeding and this year (1939) participated in the All Russian Agricultural Fair. The committee in charge of the Shevchenko Collective Farm arranged for Tovaristch Kozorez a much deserved retirement. A new house will be built for him, and at the expense of the Collective Farm he is guaranteed produce and fuel for life."

Under the heading "The Oldest Voter," the newspaper *Caspian Commune* of December 26, 1939, reported that on December 24th in the town of Guryev, citizeness A. A. Kuznetsova voted at the Soviet elections. She was born in 1818 in the former Region of Nizhni-Novgorod.

The above leads us to the conclusion that neither one hundred years, nor even one hundred and fifty is the limit of man's life or of his ability to work.

In the fall of 1937 the Institute of Clinical Physiology of the Academy of Sciences of the Ukrainian SSR sent a small group of investigators to study the people of Sukhum. Within ten days they located in the surrounding settlements a dozen men whose ages ranged from

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one hundred and seven to one hundred and thirty-five. All these old men were well and fit and proved to be excellent hosts. Some of them climbed the trees supporting grapevines in order to get for their visitors the finest grapes. When they learned of the purpose of the expedition, they volunteered to go back to Sukhum to undergo thorough laboratory tests and thus serve mankind in its fight for longer life.

One of these men, one hundred and seven years of age, stubbornly denied his right age and insisted that he was only seventy. Confronted by his cronies and witnesses, he confessed: "I want to get married. . . . Who would want to marry a man of one hundred? But any woman will marry one of seventy . . ."

The expedition was intended merely as a preliminary investigation and lasted only a month. The study of problems of longevity, the nature of aging of an organism, and the methods of fighting for a longer life will call for many years of experimental and clinical laboratory work. It will call for the organization of special homes for old animals and clinics for studying the pathology of old age. Therefore, it is still premature to draw general conclusions based on the few observations made by the members of this expedition.

I will emphasize only one observation which is common to all reports made during the investigation. This is the uniform co-ordinated quantitative adjustment of all functions of the organism to the decreased possibilities of age. However, the qualitative side of most functions has been preserved.

Hufeland, probably, had in his mind this harmonious adjustment of an aging organism to the changed pos-

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sibilities when he asserts: "If a man, during the last period of his life when his strength begins to diminish, did not grow old, he would have ended his life journey earlier." Then he adds that in old age: "One must guard against all excesses: bleeding, strong cathartics, sexual intercourse, overheating, etc. They exhaust the small store of energy and increase the organism's dryness."

The age of eighty is often considered as an advanced age. However, we can name many outstanding people who have lived to the age of eighty and over, retaining to the end of their days their creative powers and not infrequently their youthful fervor, then dying as a result of accidental, usually infectious, disease.

Leo Tolstoy died at the age of eighty-four of pneumonia.

Goethe died in 1832, also of pneumonia, at the age of eighty-three. Only the year before he had completed the second part of *Faust*. To the day of his death Goethe preserved his creative talent, capacity for work, and his interest in women. On the very last day of his life, while delirious, he exclaimed: "Look at that beautiful woman's head with its dark curls against a black background!"

Also at the age of eighty-three (in 1885) Victor Hugo died, full of life and creative power. His tragic love for the actress Juliette Drouet is well known. When in 1883 she was dying of cancer, Hugo sent her a note: "I have loved you for fifty years; it is a most beautiful marriage." Hugo's tendency to infatuations, however, caused his faithful friend much unhappiness.

Cornell and Ibsen both died at the age of seventy-eight, and both worked and loved until the end. The

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same might be said about many great masters of art.

Titian died at the age of ninety-nine of the plague. At the age of ninety-five he finished his famous painting, *Christ Crowned with Thorns*.

Corot painted one of his masterpieces on his eightieth birthday. Michael Angelo died at the age of eighty-nine and until his death did not leave his work and hobbies, going from sculpture to architecture, from painting to writing poetry.

The famous painters, Jacobe Robusti Tintoretto and Jean Laurens, died at the age of eighty-two. Jean Baptiste Greuze died at the age of eighty; Renoir at seventy-eight. The German painter Waldeck, who died at the age of one hundred and nine, exhibited his new works in 1875 when he was one hundred and one years of age. The sculptor Rodin died at the age of seventy-seven; the composer Verdi, at eighty-eight; Rameau, at eighty-five. Auber, who at the age of eighty-seven, wrote the operetta "The Dream of Love," died at the age of ninety. Auber liked to say: "I'm not eighty. I'm only four times twenty."

The famous violin makers, Amati and Stradivarius, died at the age of ninety-eight and ninety-three respectively. None of these people ever stopped creative work.

Voltaire, who suffered ill health, used to say that he owed his longevity to love for work and to his gay disposition. In his workroom were always five desks with notes and books in preparation.

Galileo was over seventy when he discovered the diurnal and monthly librations of the moon. When the famous Soviet physiologist, I. P. Pavlov, died at the age of eighty-six of an infection, he was still full of creative

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energy. The great Russian geneticist, Mitchurin, who died at the age of eighty-one, to the last day of his life supervised the researches of his collaborators. The Cossack folk singer, Djambul, at the age of one hundred still amazed his listeners with the youthful freshness, variety, and brilliancy of his songs.

14. *Community Struggle for Normal Longevity*

THE systematic scientific research into aging and the prophylactic measures to postpone its premature onset have really only begun. It would be untrue, however, to say that there are not some theories that have a sufficient scientific basis.

Why then does the majority of men never reach even the age of one hundred years? There are numerous reasons for premature aging and death.

First, there are social reasons: hunger, cold, damp, crowded living quarters where the sun seldom penetrates; constant overwork at jobs that bring no joy; or poverty caused by unemployment. All of these sap body strength and create conditions favoring the occurrence of various diseases that lead many a man to his untimely grave.

With the removal of these social obstacles, man's chances of reaching a normal, long life and, therefore, the chances of success of the science of longevity, will naturally reach a new high.

The other group of causes of the organism's untimely wearing out is biologic. Scientists, naturalists, and physicians have always thought a great deal about these causes. During the course of thousands of years all sorts of speculations came forth, some of them curious and nonsensical, reflecting the ignorance and superstitions of the epoch, others fully rational, showing profound

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observation. These latter deserve our serious consideration.

Hufeland wrote: "In Egypt, the mother of all that is strange . . . it was thought that the way to prolong life was through vomiting and sweating. It was the custom to take an emetic at least twice a month. On meeting, friends asked one another, 'How is your sweating?'"

"With the Greeks speculation on longevity took another turn. They were very soon convinced that a sensible use of the gifts of nature and constant exercise were the best means to longevity. Hippocrates, who reached the age of one hundred and four, and all the philosophers and physicians of that time knew no better aid for a long life than temperance, use of fresh, clean air, bathing, daily massage, and physical exercise.

"The great scientists and philosophers never forgot that mental exercise should be used to the same extent as physical."

"Plutarch, who himself proved the efficacy of his teachings by living to a ripe, happy old age, taught: 'Keep the head cool and the feet warm. Instead of resorting to medicine at the slightest ailment, it is better first to abstain from food and then not to neglect the body for the sake of the spirit.'"

Hufeland, himself, one of the most popular physicians at the turn of the 18th Century, coined a few interesting aphorisms regarding doctors and medicine. He said,

 "The use of drugs aims to produce an artificial disease in order to destroy a natural one."

"One doctor is better than two; two are better than three, etc. As the number of doctors increases, the chances of getting well decrease."

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In his "Opinions," Cicero said: "One must oppose old age and avoid its limitations as one does sickness. You should take care of your health, work a little, take in only as much food and drink as is necessary to keep up strength rather than stuffing yourself. You should take care not only of the body but also of the mind and of the soul. They are like flame to which no oil is added and will immediately die out when you get old.

"Hearken, my dear youth, to the ancient tale of Arkhip of Tarent. He used to say that nature never produced any dreadful disease which could be considered a luxury made to tempt unbridled desire. Luxury turns man into a traitor to his fatherland, leads to the destruction of empires, to secret communications with the enemy. Finally, there is not any kind of lawlessness, there is no crime that a yearning for luxury does not lead to. Lechery, adultery, and all sorts of disreputable acts are caused by nothing else so much as by a desire for grand living. And as for our minds—God's highest gift to man—for this divine gift there is nothing worse than luxury.

"A lewd and uncontrolled youth surrenders his weakened body to old age.

"Old age can be happy with a calm, orderly way of life. Such was Plato's way, who at the age of eighty-one died at his writing. Such was Isocrates' way, who wrote a book at the age of ninety-four and lived five years after that. His teacher, Leontin Gorgy, lived to the age of one hundred and seven, never stopping his teaching or occupation. When he was asked why such a long life was pleasant to him, he answered: 'Because there is nothing in old age I can complain of.' "

15. *Attempts at Rejuvenation*

AT ALL times man has lived as through a final drama preceding the lowering of the curtain of life. Death often breaks in upon us like a terrible enemy carrying away its victims, still full of life.

Is it surprising that under fear of old age and death man at all times looked for means to avoid them or, at least, to postpone them, looked for means to prolong or to recapture his lost youth? It is natural that at all times man tried to rejuvenate his body. In Rome, for example, the following inscription was found on an ancient tomb:

“To Aesculapius and to Health

L. Claudius Hermipp,

who lived one hundred and fifteen years and five days with the aid of the breath of young women, to the surprise of physicians. Lead your life accordingly.”

In those ancient days it was thought that the air exhaled by little girls could help in preserving and prolonging life. The Bible tells us that King David sustained his life by means of geroconomia. This method of rejuvenating the old consisted in placing young girls in bed with them. According to Hufeland this method was recognized by a famous 18th Century physician, Boerhaave. Hufeland himself, despite his many correct

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ideas regarding the proper methods for life prolongation, with which the reader is already familiar, was far from being free from the prejudices and errors of his century.

Numerous attempts to find the elixir of eternal life were made during Medieval times, the epoch of search for the "philosopher's stone." This was the epoch of superstition and hypocrisy; the epoch of alchemists, magicians, and all sorts of charlatans.

In the 15th Century the fame of Paracelsus spread far and wide through Germany. His elixir for immortality, something like the Hoffman drops (Hoffman anodyne) did not prevent his dying at about the age of fifty. Even in the 18th Century there was no lack of charlatans. It will be sufficient to mention Count St. Germain, whose elixir for longevity consisted of a mixture of sandalwood, senna, and dill. There was the "heavenly bed" devised by Graham. This bed was of a very complicated construction in which electric current, music, and perfume were used for sexual stimulation. This project, however, soon failed and the dismantled parts of the heavenly bed were sold at auction.

Contemporary science, as we have seen, pictures the process of aging as a gradual weakening of the reactivity of the cells. This process is based on biophysical and biochemical changes of the cytoplasm, changes in its physicochemical structure, a gradual loss of the cell's ability to multiply and to renew its biochemical structural elements, and the cluttering of the cell with inert particles, produced by its own cell plasm. All this leads to the contemporary attempts of rejuvenating based on concrete physiologic premises.

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The beginning of these attempts is customarily associated with the name of Brown-Séquard. This French scientist is credited with the idea of the internal secretory function of the male sex glands and the demonstration of their rejuvenating effect. He believed that sexual excitation had a similar (rejuvenating) effect. According to Brown-Séquard this excitation stimulated the internal secretory function of the glands and increased the tone of the nervous system. However, Brown-Séquard's ideas can hardly be considered fully original. Hufeland a hundred years earlier wrote that as a result of an abuse of sex glands, "they secrete less and less life-giving fluid; consequently, less and less of it is absorbed by the blood, and eventually we suffer from this." Hufeland emphasized the importance of the old saying: Coitus modicus excitat, coitus nimius debilitat (temperate sex life stimulates man; excessive, weakens.)

Extracts proposed by Brown-Séquard did not justify his hopes for combatting old age; however, as tonics they retained a place in medicine in the various forms of sex glands preparations (spermin etc.). Many contemporary theories of rejuvenation regard old age as a result of diminution in function of the glands of internal secretion, particularly the internal secretion of the sex glands. To boost this function, Steinach suggested some thirty years ago that the spermatic cords be tied off, thereby saving the energy normally used in producing wasted sperm and thus increasing the amount of internal secretion that is passed into the blood stream. Voronov believes that he has found a more radical means for rejuvenation by transplanting pieces of testes from younger men or apes into aging men.

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Recently various methods have been suggested to increase the arterial blood flow that feeds the male sex gland, in order, thereby, to stimulate production of the internal secretion of this organ. These experiments show that the substances contained in the male sex gland are capable of stimulating the biochemical processes of the body. But these new methods also proved incapable of creating inside the cells new store-houses of energy for biochemical regeneration of tissues. At present it is entirely clear that the rejuvenating effect of these operations is obtained only when untimely aging is caused by an early diminution of sexual function. Most frequently, the stimulating effect is short-lived, manifesting itself in a spurt of virility that shortly gives way to even greater senility. It is very interesting to note that this spurt according to Rudziska, one of the originators of the physicochemical theory of aging, is accompanied by a characteristic change in the colloids of the tissues. Extract from tissues of rejuvenated rats have a high degree of colloidal dispersion equal to that of extract from young rats. In other words, from the physicochemical point of view rejuvenation has taken place.

Naturally these experiments in rejuvenation have aroused universal interest and, for this reason, we must dwell on them for a moment. The beneficial effect of transplantation is attributed to a continuation, for a few weeks, of the secretory function of the transplant. Subsequently it disintegrates and liberates a certain amount of hormone which passes into the blood. However, the most plausible explanation is that the products

of disintegration of transplanted cells stimulate corresponding cells in the organism.

In the last decades a very interesting biologic law has been taking shape. When cells disintegrate, substances of a rather complex structure are formed. Carrel called them proteoses. They act on analogous cells, stimulating their functions—nutrition, growth and multiplication, i.e. regeneration. In other words, the products of the tissue's disintegration act like hormones or more correctly according to Drs. Medvedeva and Bogomolets as autocatalysts,¹ because it is hard to believe that they have a specific chemical action that is characteristic of hormones.

Haberlandt demonstrated the formation of these substances from the disintegration of tissues in wounds. Under their influence the healing of wounds is accelerated. Haberlandt called them "hormones of the wounds." Caspari found them produced at the initial stages of putrefaction of dead tissue and called them necrohormones. Spurred by the theory of autocatalysis, Academician Filatov successfully grafted skin from a cadaver on patients whose skin of the face and hands had been affected by tuberculosis and other diseases. The transplant produced autocatalysts which stimulated the nearest cells of the reticulo-endothelial system intensifying their reactions to antigens and aiding in the process of healing. Embryonic tissues are particularly rich in catalyzers, i.e. substances that stimulate cell multiplication.

These facts concerning the phenomena of autocatal-

¹ Autocatalysts—enzymes that digest or otherwise modify cells similar to the cells that produced them.

ysis will serve us later when we discuss the still little known, but promising method of cytotoxic stimulation of the functions of the organs.

Romeice succeeded in rejuvenating old rats by implanting pieces of liver (not sex glands) taken from young rats. This observation supports the hypothesis that, in the case of transplantation of sex glands, the phenomenon of the autocatalysis has also an important role.

Many experiments have been conducted involving the transplantation of other endocrine glands for the purpose of rejuvenation. Pende especially recommended a simultaneous transplanting of pieces of three glands viz., thyroid, pituitary, and testicle, because they are particularly closely related to growth regulation.

But even if there is no doubt that untimely aging and wearing out of the organism may be caused by a diminution of the function of the pituitary gland, the thyroid gland, and the sex glands, it does not mean that old age is always brought about in this manner. On the contrary, even the glands of internal secretion, like all other tissues of the body, follow the common law of gradually losing the ability of their cells to regenerate. In the preceding pages the reader has become acquainted with a series of various causes that are responsible for the loss of this ability. The problem of preventing untimely aging is to remove these causes. But medicine must concentrate its efforts on maintaining a balance in functions of the organism and on retaining these functions at a normal (neither excessive, nor subnormal) level.

I have already shown that old age prophylaxis—the

fight against its premature arrival—must begin before the organism is conceived, continue while it is in its mother's womb and then throughout life.

The question often asked is: "When does old age begin?" In this connection Rolleston wrote:

"With women the discontinuation of menstruation is the old age border line. It was said that between the ages of fifty and sixty an analogous transformation takes place in the male, as a result of changes in the sex glands. This opinion is without proof. To assert that old age begins at fifty or even at sixty is to invite an argument that this is nothing more than the middle age period."

It is not an accident that the old Russian saying has it: "Gray hairs in the beard, the devil in the ribs." Contemporary medicine has introduced the term "critical age." The French doctors consider it the beginning of the autumn of life (*Automnose*). They talk of the "diathesis of the fifties"—that is, of a special predisposition at that age to certain diseases. At that age men very often suffer from cancer, high blood pressure that leads to untimely failing of the heart and kidneys, and to cerebral hemorrhages. It is possible that one of the causes of these deep perturbations in the organism is some qualitative change in the function of the internal secretion of the sex glands. This very serious problem has not been sufficiently studied as yet. The man who safely lives through this critical age has many chances to live to very old age. However, disturbances in the internal secretions of the sex glands does not mean that old age has arrived.

And even with regard to women, Rolleston's opinion

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is not correct. Statistics show that men do not live longer than women, although the ovaries often cease functioning during the first half of life when women are still comparatively young.

The newspaper *Izvestia* of January 4, 1939, reported from Irkutsk: "In the village of the Katansky District lives an experienced hunter; an eighty-five year old citizeness Anna Vasilyevna Mnogo. Although the hunting season has just started, she has already bagged 160 squirrels, whereas the young hunters have killed only between 50 and 70 squirrels. Citizeness Mnogo has been awarded the badge: 'The outstanding hunter.'"

Statistics repeatedly have shown that the number of long-lived women is greater than that of men. *Pravda* of September 30, 1939, reported the results of the last census: "On September 17, this year, in Moscow, there were 55 persons older than 100 years (6 men and 49 women), and 611 persons between the ages of 90 and 100, (91 men and 520 women)."

Many prematurely aged men are often sexually virile, and infrequently it is through the abuse of this function that the general senility is brought about.

An occasionally observed abnormal sexual maturity in children, accompanied by an excessive influence of the hormones of the sex glands, results in a rapid aging of the person. In some such instances, castration has a rejuvenating effect. According to the observations of a number of authors, castrated men and animals have a normal life span.

In literature the question is frequently aired regarding the influence of sex on man's creative ability. Unquestionably this influence does exist.

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It would be a mistake to consider the aging of the organism as a result of exhaustion of the endocrine glands, particularly the sex glands. The significance of the latter as a rejuvenating factor that keeps up the tone of the organism, although often highly exaggerated, is definitely beyond question. The dwindling away of the sexual function, however, is more often the sign of old age, rather than its cause. The proof of this may be seen in the disillusionment which replaced the enthusiasm for rejuvenation by means of transplanted sex glands.

The above is not intended to detract from the importance of the endocrine glands for man's longevity. We know well what profound changes in the organism and what serious diseases are connected with disturbances of their functions. We know that without many of them our very existence would be impossible. Some other causes may, however, play an equally important part in an untimely withering away of the organism. They are exhaustion of physiologic energy, and exhaustion of energy needed for the biochemical regeneration of cells of the heart, nervous system, reticulo-endothelial system, etc.

The search for methods of preserving the ability of the cells to regenerate is one of the primary problems in the fight against untimely aging. In this connection autocatalysis requires particular attention. When sex glands are transplanted for the purpose of rejuvenation it is autocatalysis that stimulates the dwindling function of the analogous glands in the recipient. The products of disintegration of the transplant in some

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cases succeed in sustaining the regenerative and functional properties of the various organs which have shown signs of decline in activity. This question deserves the most careful experimental and clinical study.

16. *Blood Transfusion in the Fight for Longevity*

IN THE dim distance of past epochs, in ancient Hellas, a myth was created about an enchantress, Medea, who was able to bring back life and youth by blood transfusion.

During the Middle Ages, the idea of rejuvenation by blood transfusion was colored by the superstition and sorcery typical of that epoch. More than once it was a cause of shocking crimes when an attempt to rejuvenate an old man with the blood of an infant would result in the death of both of them. In the Renaissance, this idea shared the questionable reputations attributed to scholastic ravings, and partook of the odors of miasmas that came forth from the ruins of the Middle Ages.

Harvey's discovery of blood circulation, some three hundred years ago, again aroused interest in blood transfusion. The frequent failures, however, that often cost the patient's life, caused a discontinuance of this operation, despite the beneficial effect obtained in many cases of blood transfusion. A new era in blood transfusion dawned with the present epoch, when in the beginning of this century Landsteiner presented the method of typing the bloods of donor and recipient.

It was discovered that the fluid part of the blood (plasma or serum) of some people possesses the ability to agglutinate or clump the red blood corpuscles of

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others. This is the explanation of failures occurring in transfusions. The red blood corpuscles of the donor, when agglutinated, plug vital blood vessels viz., vessels that feed the nerve centers, or cause obstructions of vessels in the kidneys thus stopping the formation of urine, as a result of which the patients die of uremic poisoning. Landsteiner, his pupils and his followers, showed that all men can be divided into four groups in accordance with the presence in the blood of agglutinins, substances causing agglutination.

The corpuscles of the first group (Group O) are not agglutinated by any serum. That is why the blood of this group can be given to any person, regardless of the group to which he belongs. This is the most common group (about 40%) and has been named the group of universal donors. On the other hand, the red blood corpuscles of the rare fourth group can be agglutinated by all the other groups. For this reason the blood of a person of the fourth group can only be transfused into a patient belonging to this group. Blood plasma of the fourth group contains no agglutinins [i.e. it is unable to agglutinate any corpuscles—Transl.]. Therefore, a man of the fourth group can safely receive blood of any group. The second group agglutinates the red blood corpuscles of the third and fourth group. Therefore, in the second group only the blood of the second and first groups can be used for transfusion. The third group agglutinates the red blood corpuscles of the second and fourth groups, therefore it can utilise only blood of the third and first group. Finally, the first group agglutinates the red blood corpuscles of all the

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other groups. Hence, with the first group only the blood of the first group can be used.

When, thanks to discovery of blood groups it became possible to determine the compatibility of the bloods of the donor and the recipient before transfusion, this operation proved to be entirely safe and gained wide use, especially after the First World War. During that war Americans resorted to blood transfusion on a large scale. In the Czar's armies it was not used. At the present time the USSR probably stands first in the organization and promotion of blood transfusion.

Statistics show that up to 40% of the wounded, who die on the battle field, could be saved by a timely blood transfusion. In instances of heavy losses of blood from wounds or during shock caused by a wound it is often the only means of lifesaving. During transfusion the deathly pallid skin becomes pink, breathing and pulse are restored, the motionless, half-dead man opens his eyes, consciousness returns—a life has been saved.

But where can one look for a donor under battle conditions? Moreover the process of transfusing blood directly from one man to another is a rather complicated procedure. It was essential to develop a method of blood preservation. This problem was completely solved many years ago by the Central Moscow Blood Transfusion Institute. Blood prepared in Moscow was carried by rail to the farthest corner of the USSR, and there successfully transfused to the sick. Preserved blood is good for fifteen days. This period can be raised to even a month. At the present time each large hospital, each lying-in hospital, has a ready blood bank of various blood groups instead of "on-duty" donors. To give this

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blood to a patient is a matter of a few minutes. The last International Congress for Blood Transfusion recognized that in war only blood banks could be widely used.

In wartime, great numbers of blood donors are required for preparation of these blood banks. Each healthy citizen must become a blood donor. It is easy to see that the loss of 500 cubic centimeters of blood does no harm to the healthy person.

But what has this to do with longevity, with the struggle against old age, the reader might rightfully ask.

To understand the role that blood transfusion can play in preventing untimely aging of the organism, it is necessary to analyze the mechanism by which this transfused blood acts. Here I shall refer largely to investigations of my own and those of my numerous collaborators at the Central Moscow Blood Transfusion Institute and also at the Kiev Institute of Experimental Biology and Pathology which I direct (Nina Medvedeva, Fedorov, Yudina, Oleg Bogomolets, and others).

Transfused blood has two effects, that of a substitute and that of a stimulant.

The mechanism of blood substitution really needs no explanation. When the organism lacks its own blood, due to either heavy hemorrhage or serious anemia, transfusion supplies this lack. The transfused red blood cells are retained by the recipient for a long time. We may say, two months. The plasma proteins are retained for a considerably shorter time. Of still shorter duration is the effectiveness of the enzymes and hormones from the transfused blood. There is therefore no reason to

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expect a lasting rejuvenating effect in an old man from blood transfusion, even if all of his blood is replaced by that of a young man.

The substituted blood, however, has also a stimulating effect which strengthens all functions of the organism. To explain this reaction I proposed fifteen years ago the theory of colloidoclastic shock, which apparently is gradually gaining general recognition. This theory can be explained very simply.

To begin with, what is the meaning of the name itself? Shock means a blow, an impact. Colloidoclastic means damaging the colloids, in this case the particles of albumin that enter into the composition of the blood and cytoplasm.

Professor Medvedeva has shown that during transfusion of blood, even when the recipient and donor both belong to the same blood group, colloidoclasia of the cellular proteins of the recipient always occurs when these meet the albumin of the blood plasma of the donor. As Oleg Bogomolets' researches proved, during the colloidoclasia that occurs in a transfusion, complex electric processes take place, a sort of electric storm, between the protein particles of the donor and those of the recipient. In consequence the electric tension of the surface of many particles is lowered. They become less stable, clump together and form a precipitate. Then, as Medvedeva has shown, this precipitate which consists of the older, less active particles, disintegrates.

Thus damage to the cellular colloids, colloidoclasia, causes precipitation, within the cells, of the more used particles and hastens their consequent dissolution. This enables the cells to free themselves of old elements in

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the cytoplasm. When these elements disintegrate, substances apparently are formed that resemble the already mentioned autocatalyzers which intensify metabolism and other functions of the cells. As we can see, it is not only stimulation but rejuvenation of the cell that follows the colloidoclastic shock caused by blood transfusion.

Acting on my theory of colloidoclastic shock, the lecturer Vitaly Arkhangelsky used a blood transfusion in a young patient whose vitreous humor (the jellylike, transparent substance filling the eyeball behind the lens) became opaque and for whom the usual methods of treatment had failed to restore sight. A few weeks after the transfusion he was able to see again and to return to his job. This method of treating opacity of the vitreous humor has since been tried successfully many times, both by Arkhangelsky and by other ophthalmologists. It deserves our interest not only as a successful application of theory to practice, but because it enables us to see the process of colloidoclasia with our own eyes. Shortly after the transfusion, flakes begin to appear in the heretofore evenly opaque vitreous body and the vitreous humor becomes translucent. Then the flakes dissolve and disappear without a trace.

In blood transfusion we have a powerful physiological activator that strengthens the functions of the organism. This activator has been far from adequately utilized in contemporary medicine.

Professor Ziukov was successful in employing the colloidoclastic reactions in blood transfusions for treatment of scarlet fever and dysentery. In the Institute which I direct, Professor Medvedeva showed that blood

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transfusion strengthens the ability of blood plasma and phagocytes to destroy microbes. Blood transfusion eliminates the increased sensitiveness of the organism, restores and strengthens its normal reactions and, when used at the very beginning of an infectious disease, may bring about a quick recovery.

The stimulating effect of blood transfusion spreads throughout the cells of the organism and, consequently, to the cells of the reticulo-endothelial system.

I have already shown how important this system is in preventing the development of cancer. At present the main method of cancer treatment is surgery. But after the operation small carcinomatous nodules often remain from which a tumor again develops. To combat the recurring cancer it is necessary to resort to repeated blood transfusions to stimulate the cells of the connective tissue to destroy the remnants of the tumor.

I would like very much to advise that, during the first month after a cancer operation, the patient be given three small blood transfusions (of 200 cubic centimeters each) and that this be repeated about three times during the year. There is every reason to believe that in this way it will be possible to reduce considerably the recurrence of cancer after an operation.

Cancer belongs to that class of common diseases, the fight against which is the fight for longevity itself. Besides its effect on cancer, we must study the influence of blood transfusion on the prolongation of life.

17. *The Cytotoxic Stimulation of Bodily Functions*

AT THE very beginning of this century Metchnikoff wrote: "The fight against pathological old age should consist, on the one hand, in strengthening the more valuable elements of the organism and, on the other, in weakening the aggressiveness of the phagocytes.

"This problem is not solved, but its solution does not represent anything unattainable. It is a scientific problem like any other. The properties of cells change easily under various influences. Therefore, there is nothing unreasonable in seeking methods to strengthen the blood corpuscles; the nerve, liver and kidney cells; and the fibers of the heart and other muscles. This problem is made easier by the discovery of serums that affect these various elements.

"The principle of preparing these serums is as follows: specific cells, red blood corpuscles or spermatozoa or kidney or liver cells are injected into an animal of a different species. After several injections the serum of the animal becomes active in relation to the cells that have been injected into its body. Bordet discovered these serums. The serums are called cytotoxic, i.e. toxic for various cells.

"It has been proven that small doses of cytotoxic serums, instead of killing or dissolving specific elements of tissues, strengthen them. This is similar to a reaction

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observed in connection with many poisons, namely, large doses kill, whereas small doses cure and improve the condition of certain elements of the organism. Following this rule, it is possible to show that serum which possesses the power of dissolving red blood corpuscles of human blood, actually increases their number, when injected in small doses. Likewise serum which destroys white corpuscles of a rabbit, actually strengthens them, when injected in sufficiently small quantities.

"And so here is the road to follow in order to strengthen the 'noble' elements of the human organism and to prevent a man's growing old. One might think that this problem is easily solved. All one should have to do is to inject into horses some thoroughly ground human organs, such as the brain, the heart, liver, kidneys, etc. in order to obtain, after a few weeks, serums that would affect all these organs.

"In reality the problem is much harder. Therefore, one need not be surprised if attempts to strengthen the 'noble' elements of the human organism will require a very long time."

The basic problem, which Metchnikoff pointed out, consists in "the difficulty, while testing the effects of various serums, of determining the dosage of the cytotoxic serums." This difficulty evidently compelled Metchnikoff and his collaborator, Bezredka, after publishing their experiments in 1900, to drop this investigation.

Thirty-five years ago, while studying the functions of the adrenals, I myself became convinced that with small doses of cytotoxic serum it was possible to intensify considerably the endocrine activity of these glands. But I, too, at that time encountered insur-

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mountable difficulties in establishing the dosage of the serum.

Only in 1915 was I able to assay the potency of the cytotoxic serum through modifying the Bordet-Gengou method, which was developed for an entirely different purpose i.e. for diagnosing infectious diseases. Since that time my colleagues and I have completed a series of experimental and clinical researches which, it seems to me, are not devoid of interest from the point of view of the struggle for longevity. I shall present briefly a few of the results.

Having prepared a cytotoxic serum, acting upon the reticulo-endothelial system (anti-reticular serum), we undertook the study of its effect upon the formation of substances in the organism that combat infection. These experiments proved that the reticulo-endothelial system produces substances that combat infection and that the anti-reticular serum, in stimulating doses, does increase the resistance of the organism to infection, by intensifying the activity of the cells of connective tissue.

My co-worker Joseph Neiman and I applied cytotoxic stimulation to connective tissue in cases of cancer. It has been observed that this stimulation considerably reduces the percentage of successful transplantations of cancer to rats, because the stimulated connective tissue quickly destroys the transplanted particles. Later Neiman caused reabsorption of large carcinoma tumors in rats by means of repeated injections of very small doses of anti-reticular serum.

Dr. Oleg Bogomolets has demonstrated, on extensive experimental and clinical material, the accelerating influence of our serum on the healing of bone fractures.

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In the course of time, after extensive experimental verification on rats, we started applying the anti-reticular cytotoxic serum to people. In Professor Ziukov's Clinic for Infectious Diseases, Dr. Marchuk and I tried the action of small .3 to .5 cubic centimeter doses of anti-reticular cytotoxic serum in scarlet fever. If the serum was given on the first days of the disease, it did, in many cases, stop the disease. After a three to four hour slight rise, due to the serum, the temperature would drop to normal and within twenty-four hours the patient would be on the road to recovery.

At the same time, in the Kiev Institute of Experimental Biology and Pathology which I direct, Professor Kavetsky and his colleagues, in the department of oncology, have begun studying the action of anti-reticular cytotoxic serum on human cancer. This study will require a long time. Nevertheless, we can already claim that, with the help of stimulating doses of anti-reticular cytotoxic serum, in the majority of cases of cancer patients, we are able to restore the property of the blood to cause dissolution of the cancer cells. This ability, present in healthy people, is usually lost forever when cancer develops. Since our serum acts specifically upon the reticulo-endothelial system, this study shows that the substances that destroy the cells of cancer are formed by the cells of that system. We have shown also that, by repeated injections of stimulating doses of anti-reticular cytotoxic serum, it is possible to cause dissolution of the lymphatic glands that are affected by cancer.

This is far from being the solution of the problem of cancer therapy. As heretofore, cancer has to be removed surgically. But experiments have proved that, by

stimulating the reticulo-endothelial system with our anti-reticular cytotoxic serum, it is possible to increase the resistance of the organism to cancer. There is a basis for hope that our serum will prove useful in the fight against recurrence of cancer growth after an operation. We already have data showing that the application of anti-reticular serum considerably reduces the percentage of recurrence of cancer of the stomach. (Statistics cover three years after operation.) If we consider that annually, the world over, many hundreds of thousands of people die of cancer, the prevention of recurrences of cancerous growth, at least in part of the cases, by means of our serum will represent some achievement in the fight for the prolongation of man's life. If widely used, this method may save the lives of several hundreds of thousands of men. We also hope that further improvements in the technic of preparing the serum and its therapeutic application, which are under way in our Institute, will permit us to attain still further success in the struggle against cancer.

I have already shown that my point of view regarding the importance of the activity of the reticulo-endothelial system for longevity is directly opposite to that of Metchnikoff. I consider that the reticulo-endothelial system performs a very important trophic function and that its condition, to a large degree, determines the condition of other cells of the body, not excluding the nerve cells, the centers of psychic life. For this reason, I consented to try stimulating doses of anti-reticular cytotoxic serum on a very serious and common form of psychosis called schizophrenia. According to the reports of the psychiatrists, Professors V. P. Protopopov and

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Frumkin, this serum produced a good effect in several grave cases of schizophrenia. It helped clear the mind of the patients and in many cases caused remission, permitting the patients to return home and even to resume work. Favorable results from treatment of schizophrenic patients with our serum were also obtained by the Moscow psychiatrist, Professor Guiliarovsky. These experiments started only recently. For this reason I limit myself to these few guarded preliminary remarks.

A striking effect with the anti-reticular cytotoxic serum was recently obtained by my co-worker, Oleg Bogomolets, in experiments with fractures of rabbits' extremities. Large doses of serum paralyzed the cells of the fibrous connective tissue that form the callus knitting together the ends of broken bones. On the other hand, small stimulating doses of serum (for a rabbit .001 cubic centimeter of serum) hastened considerably the process of healing the fracture and caused formation of a large bone callus. In man this accelerating action of the serum was particularly noticeable in cases of prolonged nonunion of fractures.

The reader need not be surprised at the seeming variety of the actions of anti-reticular cytotoxic serum which has proved effective in our researches with infections, cancer, and fractures. Varied and very important for the health and longevity of the organism are the functions of the reticulo-endothelial system. Our serum, however, possesses only one property, in large doses it suppresses, in small doses it stimulates the activity of this system.

If it should prove that the small doses of the anti-reticular cytotoxic serum, on repeated introduction into

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the body are able to prevent untimely sclerosis of the reticulo-endothelial system, this serum will become a very valuable aid in the struggle for prolongation of life. We have begun to study this problem. Here a great deal of circumspection and long experimentation first on animals and later on human beings is needed.

Strong and specific stimulating effects of the cytotoxic serum on the reticulo-endothelial system should be considered as well established. But it is also possible that, through a prolonged cytotoxic stimulation, the connective tissue will excessively proliferate and then become sclerotic. The question of proper dosage will be of decisive importance.

The wide use of my anti-reticular cytotoxic serum (ACS) for therapeutic purposes in the USSR has aroused great interest in this new method of pathogenetic therapy, in both the United States and Great Britain. Because inadequate knowledge of the indications and contraindications for the use of ACS may lead to errors during treatment of the sick, I believe it is essential to give at this point some brief directions.

Conferences were held in Kiev in 1940, in Ufa in 1942, and in Moscow in 1944 on the problem of the therapeutic action of ACS. From the reports made at these conferences and from the many reports which appeared in our medical journals, the following conclusions may be drawn:

ACS is a specific factor acting upon the reticulo-endothelial system. Subcutaneous injection of ACS in doses of .05 to .1 cc i.e. 1-2 drops of undiluted serum (strength of serum 1:100), has a strong stimulating effect on the cells of the reticulo-endothelial system in adults.

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Before injecting ACS it should be diluted 1:10 with sterile .85% solution of NaCl at 37° C. ACS is injected subcutaneously three times at intervals of two to three days.

Since the reticulo-endothelial system has important nutritive, plastic and protective functions, it must be concluded that the therapeutic use of ACS is indicated in all those diseases that cause depression in the above mentioned functions of this system.

The healing of fractures is often delayed for a long time on account of suppression of the plastic function of the connective tissue. Under the influence of ACS such fractures often knit together in a normal period. It is understood that there should be no pseudarthrosis present and that the bone fragments should be properly set. Thanks to the application of ACS during this war, many wounded with serious fractures have been saved from becoming invalids and have been returned to the front.

The use of ACS is also indicated in cases of wounds and sores that have not been able to heal for a long time. Delay in healing denotes a disturbance in the plastic, nutritive and defense functions of the connective tissue. In many cases normal functions are restored under the influence of ACS.

ACS is also recommended in many acute infections. An indispensable condition for success of the treatment is the earliest possible administration of ACS during the acute, allergic period of the disease. Under the influence of ACS, typhus, scarlet fever, tularemia, and brucellosis, have a milder course and shorter duration.

A great deal of interest has been aroused by the ques-

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tion regarding a therapeutic action of ACS in rheumatism. Academician Nicholas Strazhesko, in whose clinic the effect of ACS on rheumatism was studied, wrote: "ACS in stimulating doses gives the best results in the second (allergic) stage of infectious diseases. Take for example acute rheumatism. If during this period you inject the serum, you will give a stimulus to the entire mesenchyme for a final victory over the infection, truly, sometimes a single injection is sufficient to stop the acute rheumatic process. (From the *Symposium of The Therapeutic Action of ACS* page 102, Edition of the Academy of Sciences, USSR, 1942.)

On the other hand, I would like to emphasize that there is no basis for the use of ACS in chronic rheumatism. For chronic rheumatism and endocarditis ACS should not be administered.

Good results may be obtained in many cases of lung abscesses and pleural empyema (pus in chest cavity).

Gynecologists call attention to a very effective prophylactic action of ACS in cases of infection after birth and abortion. But ACS fails in cases of general puerperal sepsis (blood poisoning occurring at child birth).

ACS is recommended for general and local suppurative infections, particularly in the primary stages of the disease. In cases of grave generalized atypical sepsis of traumatic origin, ACS is usually ineffective.

Neuropathologists believe that ACS should be used for certain infectious diseases of the nervous system.

Positive results have been reported regarding treatment of gonorrhea in men and women with sulfa drugs, after a preliminary stimulation of the protective function of the reticulo-endothelial system with ACS.

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In my opinion a further study of treatment of such diseases as acute rheumatism, syphilis, and malaria by means of simultaneous administrations of drugs and ACS deserves very serious consideration. The use of ACS in tuberculosis has been studied very little and requires great caution.

The use of ACS in cancer should also be studied further. In the struggle of the organism against the development of cancer, the reticulo-endothelial system plays a big role. I consider that cancer cannot develop in a body whose connective tissue has retained normal resistance. The presence of cancer denotes a great impairment of the protective and plastic functions of this system. However, the only way to cure cancer is to diagnose it as early as possible and then to remove it surgically. ACS does not cure cancer that has reached a clinically recognizable, serious stage of development. ACS is not indicated in such cases.

Nevertheless, I would recommend, after the surgical removal of a tumor, a series of injections of ACS, which should be repeated every 2 or 3 months. The restoration of the functions of the reticulo-endothelial system, by means of ACS, aids in the destruction of the tiny metastases and prevents the recurrence of the cancer.

18. *The Prevention of Untimely Aging*

CONTEMPORARY medicine is not looking, of course, for an elixir for long life. Its problem is to find measures that will energize the cells to biochemical regeneration, will mobilize the organism's own forces, will aid in restoring normal metabolism, and will assure the continuous preservation and smooth co-ordination of the physiological systems. This problem is far from solved. But from the examples cited, the reader may be convinced that it can be solved. It is essential, however, to work hard, with faith in the creative power of scientific knowledge.

Medical science faces two new problems, i.e. active prophylaxis and the treatment of old age. In the realm of dietetics there is sufficient data for the preparation of a rational, preventive diet. The old idea of the famous Russian scientist, Metchnikoff, regarding the significance of controlling the bacteria in the large intestines is far from being fully utilized in practical life. And yet it may be a source of great benefit.

A new science is now in process of formation, a science of biochemical, autocatalytic self-regulation of functions. It opens wide perspectives for discovering methods of stimulation of cellular functions, which would intensify their ability for regeneration without causing untimely exhaustion. Some of these methods may be blood transfusion, cytotoxins, etc.

Hope for the prolongation of life should not be

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pinned on attempts to rejuvenate an organism that is old. It is hard to turn back the river current. But to slow down the process of exhaustion of the organism's functions, the process of aging, is possible by the wise control of one's life.

The first principle of a wise life is work. The entire body must work. All its functions must work. Not one of them must be forgotten. Yet, none must be overburdened to depletion. The misuse of any function at all—overeating, sexual excesses, overwork—will inevitably lead to untimely old age. Rest from work must precede fatigue. It should be used to prevent rather than cure fatigue.

It is very important to pay attention to one's breathing. One must breathe deeply, so that exhalable gases will not be delayed in the lungs, and so that the blood will be fully saturated with oxygen. One should remember that oxygen is an important nutrient and a substance that, by a process of oxidation, helps the organism not only to produce heat but also to get rid of the harmful, poisonous waste products of metabolism.

Gueniot calls the lungs "the true fountain of youth where billions of messengers from the tissues—the red blood corpuscles—rush in for the agent of life—oxygen."

Very important for longevity is proper digestion. Food should be wholesome and simple. Many doctors recommend a vegetable diet to older people. However, such a diet is far from being popular. The famous author of *In Praise of Folly*, Erasmus of Rotterdam said: "At heart I am a Catholic, but my stomach is that of a Protestant." He begged the Pope to absolve him from the necessity of observing the days of fasting. However,

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there is no doubt that one should avoid too much meat. Proteins are necessary for the replacement of the proteins of the cytoplasm. But they are needed daily for this purpose in only small quantities. Although meat is predominantly protein foodstuff, it is also responsible for the formation of waste products that are not only unwholesome but even harmful when functions of the liver and kidneys are impaired. For this reason meat should be used in small quantities.

To replace expended energy it is better to use fats and carbohydrates (butter, bread, vegetables, sugar, etc.) that are oxidized in the organism to carbon dioxide and water. It is very harmful to overeat. I will remind you of a French saying: "To get fat, is to get old."

The bowels should be emptied not less than once every twenty-four hours. This is very important. It is necessary to have proper bowel movements to prevent constipation. A glass of sour milk or a bottle of kefir at night, at the same time regulates the emptying of the bowels and suppresses the harmful bacilli of putrefaction.

Exercise and massage are useful in preventing blood congestion in various parts of the body. One should begin and end the day with them. Ten to twenty minutes daily, not only will add vigor for the rest of the day, but will preserve more than a few years of life. This promotes the exchange of substances between the tissues and the blood, improves the nutrition of the cells, and facilitates the passage of wastes from cells into the blood and from the blood through the kidneys into the urine. Walking for not less than an hour a day is a "must" for those who lead a sedentary life. One should

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dress lightly but according to the weather. Even Plutarch advised that for the preservation of good health one must keep the head cool and the feet warm, and not be in a hurry to take medicine.

It is essential to keep the skin clean because the skin is an important organ of exchange.

Sleep is important. Man should sleep seven to eight hours daily, including one hour after the main heavy meal. Sleep brings rest to all of the body's functions, particularly those of the nervous system. Overtiring this system is very harmful to the general welfare of the organism and, if it is prolonged, will lead to exhaustion, untimely aging and death. Excessive indulgence in drink and tobacco affects the nervous system very badly.

If the sex function is not overtaxed it will last longer and will favor longevity. Changing the sex function into a source of excessive enjoyment will bring on an early exhaustion of the organism and its untimely aging.

The fundamental precept of the fight for longevity is avoidance of satiation. One must not lose desires. They are mighty stimulants to creativeness, to love, and to long life.

And so we see that the ability to lengthen life is, first of all, the ability not to shorten it.

There should be no difference in the lives of those who work with their brains and those who work with their hands. Scholars must not forget their muscles and blood circulation. It would be just as harmful for men engaged in physical work to lose their interest in science, art, and artistic creativeness.

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That is why the Soviet government in taking care of its greatest treasure, man, builds palaces of labor, clubs, and parks of culture and rest, to stress the great significance of *physical culture* in connection with *spiritual culture*. The harmonious development of the body and all-round systematic care of it are the best methods of achieving a normal, long life.

Normal longevity at the present level of human development may be scientifically determined as being 125 to 150 years. There is no reason, however, to consider even these figures as limits.

THE END

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